NameRank: Measuring LLM-Mediated Recognition as the Post-Bibliometric Impact Channel

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Abstract

What a frontier model says about a person or tool is often the first description a human sees, so whether a name reached training corpora is a measurement problem. Citations explain a third of whether a model recognizes a researcher [Li, 2026]; we take the residual as our target: **NameRank**, a continuous $[0, 1]$ recognition score. We probe **[TBD:** N**]** entities in **[TBD:** K**]** cohorts with one open-ended question across 36 models; an independent judge issues a *recognition* verdict per response against a curated gold—did it state a non-guessable fact true of this exact entity?—so fluent hallucinations and context echo earn nothing. NameRank is the panel fraction that recognizes the entity. Six findings: **(i) a credential treadmill**—seven of nine Olympic-style credentials score at or below a working-researcher baseline; marquee prizes (Nobel, Turing, Fields) invert this, accruing for decades; **(ii) artifact over creator**—in 8 of 11 verified pairs the tool out-ranks its maker; **(iii)** h**-index carries the bibliometric signal**—citations add nothing once it is known, yet ∼60% of variance is unexplained; **(iv) a corpus-density gradient**—Stanford CS faculty score twice Tsinghua's, part of an English-corpus ladder across institutions and countries; **(v) external calibration**—on 258 news events with pageview ledgers, recognition loads on peak salience, not persistence; **(vi) self-report is not measurement**—ranking their own recognition, models recover the panel ordering partially ($\rho \approx 0.5$–0.6), though rarely claiming a fictional name. It stays silent below threshold, not misleading; findings survive paraphrase, cutoff, cross-judge, and confound controls. We release all probes, golds, responses, and code.

Code: `https://github.com/19PINE-AI/namerank`
Website: `https://01.me/research/namerank`

1 Introduction

The channel through which human curiosity reaches people and their work is changing. A practitioner deciding whether to read a paper, hire a candidate, attend a talk, or invest in a startup increasingly arrives at the question with whatever a frontier language model says about the entity, often before any human source intervenes. We do not measure the prevalence of this LLM-first lookup behavior—it is a usage trend we take as a premise, not a result—but conditional on it, what the model says is determined by whether the entity's name has propagated into the model's training corpus. If the corpus has absorbed the name with a clean attribution chain, the practitioner is presented with a coherent summary; if it has not, the model says "unknown" or, worse, fabricates a plausible-sounding biography. The information surface that mattered ten years ago—search rankings, conference plenaries, citation counts—has been re-mediated by a small number of frontier models, and whether a name has propagated into those models' weights is now itself a substantive measurement problem.

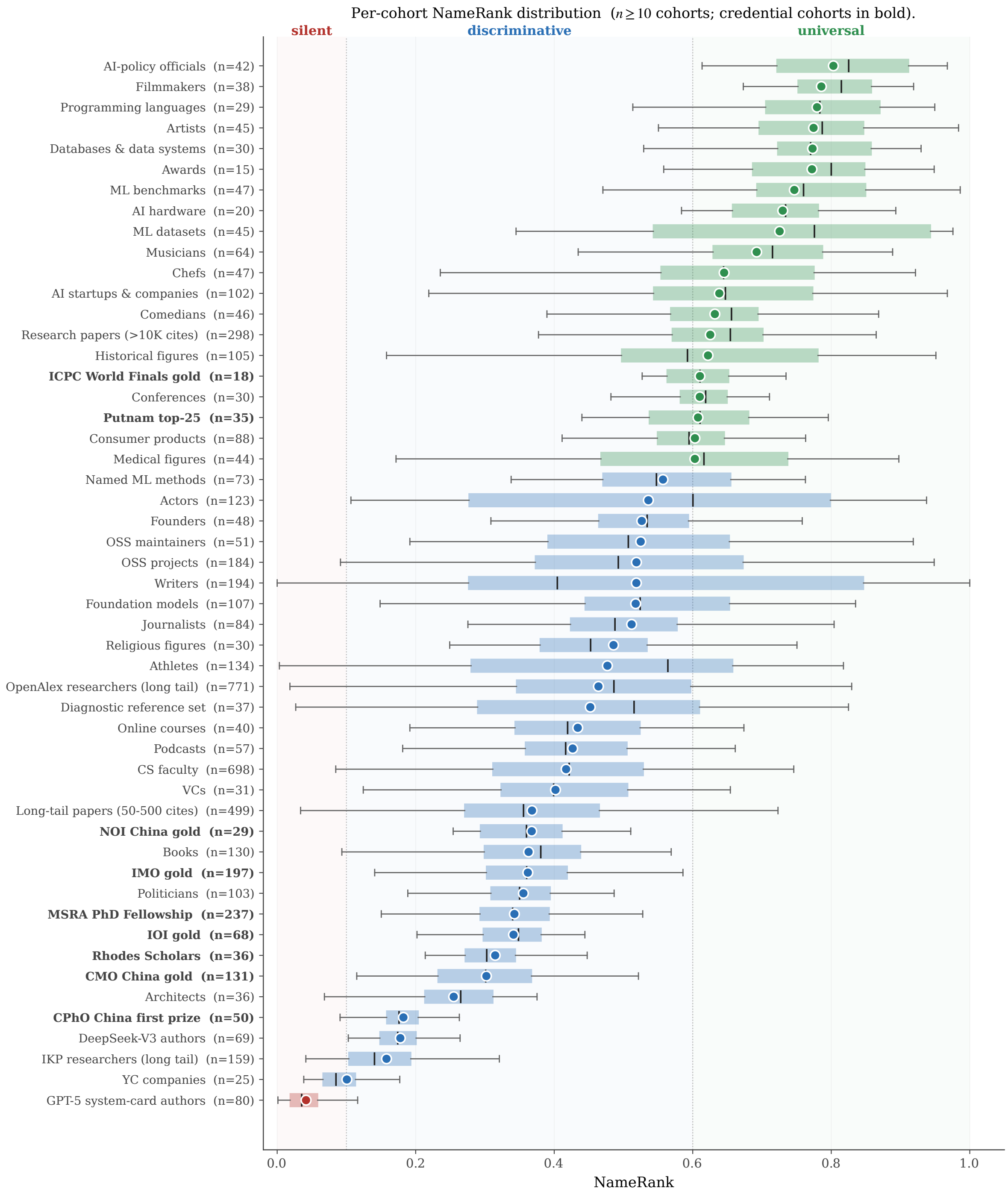

Figure 1. Per-cohort NameRank distribution. Each box is a cohort ($n \geq 10$); the dot is the cohort mean, the box covers the IQR, and the inner line is the median. Three zones are visible on the same axis: *silent* (mean ≤ 0.1), *discriminative* (0.3–0.6: working academics and credentialed individuals), and *universal* (> 0.7: named artifacts and high-profile mid-tier figures). The nine credential cohorts (bold labels) cluster in the lower half of the discriminative zone (Section 3.2).

Bibliometrics do not solve this measurement problem. In earlier work on the Incompressible Knowledge Probes (IKP) benchmark, Li [2026] found—as a byproduct of calibrating the knowledge capacity of frontier models—that citation counts explain only about a third of the variance in whether a model recognizes a researcher. A controlled audit attributed the remainder to three corpus-level mechanisms: *name uniqueness*, *named-artifact attachment*, and *subfield-ecosystem density*. That work characterized the residual but did not measure it: its recognition scores live on a coarse tier ladder built for parameter estimation, not for comparing entities. We take the opposite stance—the residual is the quantity of interest—and build a dedicated instrument for it. Section 6 details how the instrument departs from its predecessor.

A heuristic factorization makes the target precise (a framing device, not an identity we estimate—the second factor is left unmeasured):

$$\text{Recognition impact} \approx \underbrace{\text{NameRank}}_{\text{per-query presence}} \times \underbrace{\text{Query volume}}_{\text{humans querying that LLM}}$$

To the extent the query volume is approximately constant across entities at the relevant time horizon, the first factor carries the cross-entity variation, and it is what we measure. We do not claim NameRank measures *quality* or *merit*; it measures whether the corpora that mediate human curiosity at scale have internalized the entity's name.

NameRank is a continuous $[0, 1]$ recognition score. Each entity is probed with one fixed open-ended question ("tell me what you know about [name], who [context]") across a panel of 36 frontier models. An independent LLM judge reads every response against a curated gold answer and returns a binary recognition verdict—true only when the response states a specific, non-guessable fact that is correct and not derivable from the context, so a fluent hallucination or a restatement of the context contributes nothing. The per-entity score is the panel recognition rate. Three properties distinguish this construction from recognition numbers that fall out of benchmark suites: it places researchers, founders, open-source maintainers, and named artifacts on a single axis; it is hallucination-resistant by construction; and cross-model disagreement is retained as a signal rather than averaged away. We evaluate 5,719 entities across 54 cohorts spanning academia, industry, open source, and the public sphere (Section 2).

Contributions.

1. A **continuous cross-domain recognition instrument** that places researchers, founders, OSS authors, public websites, and named artifacts on one $[0, 1]$ axis—no prior instrument does (Section 3.1).
2. The **credential treadmill**: most Olympic-style intellectual credentials (IMO, IOI, CMO, Rhodes, MSRA Fellowship) score at or below a working-researcher baseline. The credential no longer propagates a name; named post-credential production does (Section 3.2).
3. The **artifact-over-creator inversion**: for independent creators, the tool out-ranks its maker. The effect is measured observationally, corroborated by a context-injection experiment, and traced to an attribution-direction asymmetry in the corpus (Section 3.4).
4. **What predicts recognition**: among bibliometrics, *h*-index carries the signal—NameRank rewards name-recurrence across distinct works rather than citation volume—though even it explains under half the variance; a corpus-density gradient runs across institutions,

countries, and prompt languages; and a 258-event news cohort with a directly recorded attention ledger calibrates the instrument externally—peak salience, not persistence, drives propagation (Sections 3.5–3.6).

5. A **validation battery** covering wording, model vintage, judge family, synthetic nulls, and gold-answer confounds (Section 4); a **self-report probe** showing that pairwise recognition introspection reads the corpus prior and cannot replace verified measurement (Section 3.8); and an **open release** of all probes, gold answers, responses, and code (links on page 1).

Figure 1 previews the central result: a single recognition axis on which every cohort in the study is placed.

2 Measuring NameRank

Figure 2 summarizes the pipeline this section unpacks: a fixed open-ended probe is sent to a 36-model panel for each entity; an independent LLM judge reads each (entity, model) response against a curated gold answer and returns a binary *recognition* verdict—did the response demonstrate genuine memory of this specific entity; and the per-entity NameRank is the fraction of the panel that recognizes it.

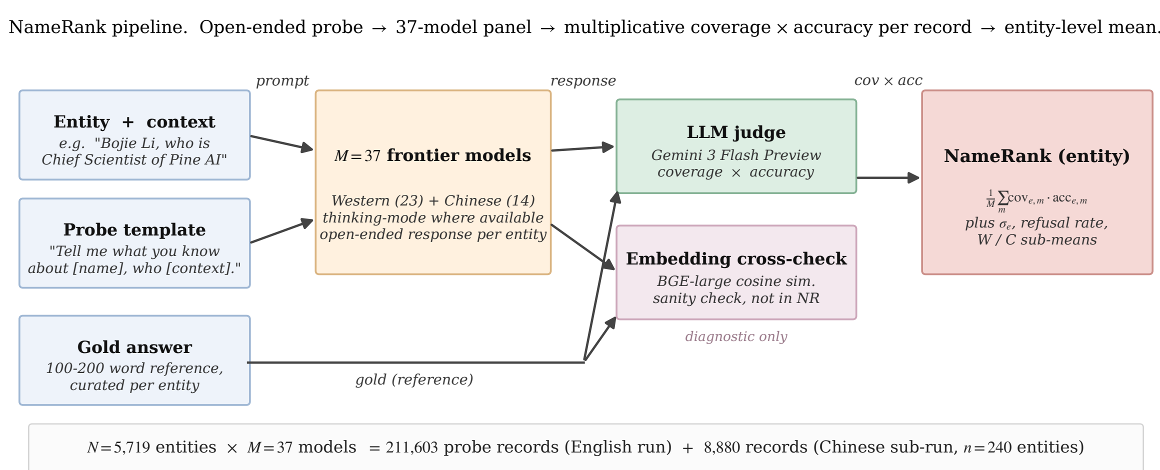

Figure 2. NameRank pipeline. The probe template is fixed, with two slots: the entity name and a short role/affiliation/field context. Each of the 36 panel models answers the open-ended probe; an LLM judge reads the response against the gold answer and returns a recognition verdict—true only if the response states at least one specific, non-guessable fact about the entity that is correct and not derivable from the context. The per-entity score is the panel recognition rate. An embedding similarity (BAAI/bge-large-en-v1.5) and a graded coverage×accuracy score are stored per record as diagnostics only—neither enters the headline score (Appendix L). Totals: **[TBD: *N* records]** over **[TBD: *N* entities]**, plus a Chinese-prompt sub-run.

2.1 Definition

For an entity e with a disambiguating context $c(e)$ and a curated gold answer $g(e)$, and a model panel $\mathcal{M}$ of size $M = 36$, NameRank is the fraction of the panel that recognizes the entity,

$$\text{NameRank}(e) \;=\; \frac{1}{M} \sum_{m \in \mathcal{M}} \text{rec}(r_{e,m},\, c(e),\, g(e)), \tag{1}$$

where $r_{e,m}$ is the response of model m to the open-ended probe

> *"Tell me what you know about [name], who [context]. If you do not recognize this entity, respond with 'unknown'. Limit your response to about 150 words."*

and rec $\in \{0, 1\}$ is a recognition verdict produced by an LLM judge (Section 2.4): it is 1 only when the response states at least one *specific, non-guessable* fact that is true of this exact entity—verified against the gold answer or the judge's own knowledge—and not derivable from the disambiguating context. A refusal, a hedged non-answer, a response confined to context-derivable generalities, and a fluent but wrong-person biography all score 0. NameRank thus reads as the probability that a randomly drawn frontier model genuinely knows the entity. Three worked (entity, model, response) triples spanning the range are provided in the repository documentation.

2.2 Probe Design

One open-ended probe per entity. Closed-factoid probes have a known failure mode: a model that recognizes the entity through fact X but does not recall fact Y scores zero on a probe-of-Y, biasing the metric toward specific-fact recall. The open-ended probe instead asks the holistic question: does the model have a retrievable representation of this entity? The template is fixed, with two slots: the canonical entity name, and a short disambiguating clause giving role, affiliation, and field (e.g., *"Wei Zhang, who is a professor of mathematics at MIT working on PDEs"*). The context resolves name collisions at probe time without revealing the gold answer. The full English and Chinese probe text and the judge prompt are reproduced verbatim in the repository documentation.

The context names role, affiliation, and field—never specific contributions, papers, dates, or named artifacts—so it anchors disambiguation without leaking the gold; the judge's rubric credits only specific named facts, and generic in-context biographies earn nothing (evidence in Appendix A; a dedicated context-ablation in Appendix Q).

2.3 Gold Answers

For each entity we construct a gold answer built from *entity-specific, verifiable facts*—named works, named contributions, real affiliations with dates, and other identifying particulars—composed from primary and authoritative sources: bibliographic records (OpenAlex, Semantic Scholar, Crossref) for researchers and papers, official competition and award records for credential holders, code-host metadata and READMEs for open-source artifacts, encyclopedia lead sections for public figures and companies, and length-normalized web-grounded profiles for the long tail. The gold serves two roles for the judge: it *identifies* which specific entity the probe refers to (disambiguating same-name individuals), and it supplies a correctness anchor. It deliberately does *not* restate the disambiguating context: a fact the probe already supplies earns no recognition credit, so the gold's discriminating content is the material a model must know independently. We audit a stratified sample against primary sources following the protocol of Li [2026]; the observed correction rate is small, typically year-of-affiliation drift. A confound check confirms NameRank does not reduce to a has-a-Wikipedia-page flag (Appendix Q).

2.4 The Judge

We use Gemini 3 Flash Preview as the judge (direct Google API to minimize routing variance; temperature 0). The judge is shown the entity name, the probe context, the gold answer, and the model's response, and returns a binary *recognized* verdict. It is instructed to credit recognition only when the response states a specific, non-guessable fact that is (a) true of this exact entity, established either by the gold answer or by the judge's own reliable knowledge; (b) not derivable from the probe context; and (c) about the same entity the gold identifies, not a same-name individual. The judge uses knowledge *asymmetrically*: it may refute freely—flagging a wrong-person biography or a false claim using anything it knows—but it may *credit* a fact only when it positively verifies it, so a plausible-sounding but unverified specific about an entity it does not recognize counts for nothing (the signature of hallucination on obscure or fictional names). For competition participants, fellowship recipients, and minor-paper authors—where an LLM judge has no reliable per-individual memory—an anti-confabulation provision restricts credit to facts present in the gold or to major, widely-documented achievements, never to fine-grained particulars asserted from the judge's own memory. Broad field, nationality, an employer inferable from the name, generic praise, and restatements of the context are all non-credit by construction.

We also record, per response, a graded coverage$\times$accuracy score and the embedding cosine similarity to the gold [Xiao et al., 2024]; both are retained as diagnostics and neither enters the headline NameRank. Appendix L shows why a judge is required—embedding similarity credits fluent hallucinations as recognized—and why the recognition verdict is preferred to a graded score: graded coverage measures overlap between two arbitrary samples of an entity's fact universe, penalizing a model that knows the entity through facts the gold happens to omit, whereas recognition asks the construct-faithful question of whether the model knows the entity at all. The dependence of the findings on the judge's model family is quantified with a multi-judge re-grade in Appendix O.

2.5 Aggregation

NameRank is the unweighted panel recognition rate across the 36-model panel (Equation 1). We also retain, per entity, the refusal rate, the Western/Chinese sub-panel rates, the diagnostic graded score, and the raw response text. Averaging across the panel is what makes the score stable: a single model contributes noise from prompt sensitivity, refusal-policy idiosyncrasies, and training-data accidents, while the panel rate integrates corpus presence over the current frontier fleet. The retained per-model verdicts keep the residual signal—an entity recognized by one vendor's models but not another's is itself informative (Section 3.5.3). A synthetic-null floor—fictional entities built on each cohort's gold recipe and scored by the same judge—calibrates the residual recognition a name that exists in no corpus can attract; scores are read against this floor (Appendix P).

2.6 Model Panel

The panel comprises 36 frontier and near-frontier models accessed via OpenRouter and direct vendor APIs, spanning Western and Chinese vendors and frontier-reasoning to small (sub-30B) classes (full roster in Appendix A, Table 2). Thinking mode is enabled wherever a thinking variant exists: for a recognition metric, thinking mode is the operational mode in which most user queries are answered, so the panel reflects deployment reality. The Western/Chinese partition is used only for the cross-vendor analysis (Section 3.5.3); the headline NameRank is the unweighted mean over all 36 models.

2.7 Entity Cohorts

We evaluate **[TBD:** *N***]** entities in **[TBD:** *K***]** cohorts spanning credentialed individuals (olympiad medalists, Putnam, ICPC, Rhodes, MSRA fellows), working academics, industry figures, named artifacts, mid-tier cross-profession figures (writers, athletes, actors, chefs, podcasts), a hand-curated diagnostic reference set (public figures, engineers, and artifacts used for case-level analyses), and special cohorts (technical-report and system-card author lists, conferences, awards, and LLM-area figures—method originators, foundational-paper authors, and best-paper-award recipients). The design goal is coverage of the axes the metric claims to unify: credentialed individuals whose recognition should be carried by the credential if credentials propagate; working academics with known bibliometrics, as the external-validity anchor; industry figures whom no bibliometric measures; named artifacts, to compare things against their makers; and mid-tier figures, to test the axis outside the technology world. The per-group breakdown is in Appendix A (Table 3); full per-cohort sizes, rates, and inclusion criteria are in the released cohort specification.

2.8 Cross-Language Sub-Run, Execution, and Release

A subset of entities—the diagnostic reference set, the Chinese olympiad and fellowship cohorts, the technical-report author cohort, and small Western control samples—was re-probed with the Chinese-translated template (gold answers and judge prompt remain English, since the judge grades across languages; cohort selection detail in Appendix I). A parallel Python runner drives the probe and judge stages. We release the probe specifications, all gold answers, all judged response records, the Chinese sub-run, and the analysis pipeline at `https://github.com/19PINE-AI/namerank` (companion website: `https://01.me/research/namerank`).

2.9 Scope: What NameRank Does and Does Not Measure

NameRank is not a quality, merit, or intrinsic-impact metric. It measures *the propagation of a specific name into the indexed corpora that frontier-model training pipelines scrape*. This correlates with academic impact through bibliometric mechanisms, with cultural reach through named-artifact attachment, and with both through derivative-content density (tutorials, blog posts, talks, podcast mentions)—but it does not measure the underlying work. The clearest counterexample in our data is a Chinese-language visa-monitoring website with roughly a million users whose NameRank is nonetheless low, because its name never co-travels with its creator's in indexable text (Section 3.7). The metric correctly registers low there, not because the artifact lacks impact, but because the metric measures something specific. We treat this as a feature: a clean instrument reports what it measures, not what users wish it measured.

Three empirical properties, each established later, govern how the score should be read. First, the axis is meaningful *across* domains: researchers, founders, OSS authors, and artifacts land on one scale (Section 3.1). Second, below the corpus-presence threshold the metric is largely *silent rather than misleading*—the panel refuses or withholds recognition instead of producing spurious rankings—against a synthetic-null floor calibrated per gold recipe (Section 3.1, Appendix P). Third, the score directly captures the residual that bibliometrics leave unexplained: for a working researcher, one widely-used, cleanly-named open-source tool moves NameRank more than one well-cited paper (Sections 3.4, 3.5).

3 Results

[TBD: This section is being regenerated on the final run. The subsections and their labels below are fixed; each will be filled with the recognition-rate results, tables, and figures. A one-line statement of the finding each will report is given as a placeholder.]

3.1 A Single Axis Across Domains

[TBD: Researchers, founders, OSS authors, and named artifacts placed on one recognition axis; the three zones (silent / discriminative / universal) and the cross-domain placement table.]

3.2 The Credential Treadmill

[TBD: Olympic-style credentials (IMO, IOI, NOI, CMO, Rhodes, MSRA) at or below the working-researcher baseline; the credential no longer propagates a name, named post-credential production does. Ladder in Figure 3.]

3.3 The Treadmill Inverts at the Marquee Tier

[TBD: Marquee late-career prizes (Nobel, Turing, Fields) clear the baseline and accrue for decades, unlike early credentials. Ladder in Figure 4; detail in Appendix C.]

3.4 Named-Artifact Amplification

[TBD: For independent creators of single artifacts, the tool out-ranks its maker. Inversion and injection in Figure 5.]

3.4.1 The artifact-over-creator inversion

[TBD: Verified (creator, artifact) pairs; the artifact out-ranks the creator in most independent-creator pairs.]

3.4.2 Attribution flows from creator to artifact, not back

[TBD: Directional mention-rate asymmetry (creator to artifact vs. artifact to creator) across the pair regimes.]

3.4.3 A worked case: Tianshou and its author

[TBD: Jiayi Weng vs. OpenAI IC peers; artifact-mediated recognition; survives translation.]

3.4.4 Context injection corroborates the mechanism

[TBD: Naming the artifact in the creator's context lifts recognition where the panel barely stores the creator; capability gradient.]

3.5 What Predicts NameRank

[TBD: Bibliometric, institutional, and geographic correlates of recognition.]

3.5.1 Among bibliometrics, h-index carries the signal—and still explains under half the variance

[TBD: h-index dominates raw citations; citations add no marginal signal; most variance remains unexplained. Regressions figure to be restored.]

3.5.2 Institution and country: the corpus-density gradient

[TBD: Top-school CS faculty recognized well above lower-corpus-density peers, even at matched citations; country gradient in Figure 6.]

3.5.3 Prompt language routes retrieval

[TBD: Prompt language routes retrieval into language-specific corpus pockets; Chinese prompts lift Chinese-named cohorts.]

3.6 News Events: The Attention Ledger

[TBD: On 258 news events with pageview ledgers, recognition tracks attention dose and loads on peak salience rather than persistence; templated series names dilute. Figure and full regressions.]

3.7 Boundary Condition: Reach Without Name Propagation

[TBD: tuixue.online—largest human reach, lowest recognition—dissociates reach from name propagation; symmetric across languages.]

3.8 Self-Reported Recognition: Asking a Model Is Not Measuring It

[TBD: Models asked to rank their own recognition recover the panel ordering only partially ($\rho \approx 0.5$–0.6); self-report reads a shared corpus prior, not the model's own state, but rarely claims a fictional name.]

[TBD: credential ladder figure]

Figure 3. **[TBD: The credential treadmill: cohort recognition rates against the working-researcher baseline.]**

[TBD: award ladder figure]

Figure 4. **[TBD: Marquee prizes clear the baseline and accrue over decades.]**

[TBD: artifact–creator inversion figure]

Figure 5. **[TBD: Named-artifact amplification: creator vs. artifact recognition and the context-injection lift.]**

[TBD: country gradient figure]

Figure 6. **[TBD: Country gradient in CS-faculty recognition.]**

4 Robustness and Validation

The findings above rest on three design choices—an LLM judge over embedding similarity, a binary recognition verdict over a graded score, and open-ended over closed-factoid probes—and on the headline numbers not being artifacts of wording, model vintage, judge family, or gold-answer construction. Table 1 summarizes the validation battery; Appendices L–Q give the full evidence. Three results carry most of the weight.

Table 1. The validation battery: each threat to the findings, the check run against it, and the outcome. Full tables and figures in the referenced appendices.

Threat	Check	Outcome	
Score reflects panel composition, not the entity	Variance decomposition over all 211,603 records; disjoint-panel re-score of the event cohort	Entity variance $\approx$3.3$\times$ model variance; eight 2026 replacement models rank events at $r = 0.905$ with the surviving panel (§3.6)	App. L
Fluent hallucinators inflate scores	Refusal and generosity structure across the panel	Accuracy axis zeroes wrong-person bios; refusal rate tracks the silent zone	App. L
A judge-free embedding score would suffice	Judge-vs-embedding comparison, 175K records	Embedding similarity is flat across judge scores and credits hallucinations	App. L
Findings are an artifact of probe wording	Four-template re-probe, 29,600 records	Rankings invariant ($r \geq 0.93$); absolute levels are template-conditional	App. M
The silent zone is corpus timing, not obscurity	Training-cutoff natural experiment	Matched difference-in-differences $\approx$0; silence is intrinsic	App. N
Findings depend on the judge's model family	Three-judge re-grade, 615 records	Judges agree ($r = 0.87$–0.93); small own-family lift moves no finding	App. O
Unknown names could score above zero	30 fictional entities through the full pipeline	Floor $\approx$0.04 (dense golds), $\approx$0.20 (boilerplate golds); adjusting for it *widens* the credential gap	App. P
Gold length, context leakage, Wikipedia presence, or web attention drive the results	Dedicated ablations and regressions; 24-month pageview baseline	All four rejected	App. Q

The scoring rule is load-bearing. Across all records, the entity factor explains far more of the variance than the model factor does, and an entirely disjoint replacement panel reproduces the event ordering—the score measures the entity, not the fleet. The refusal structure shows why a recognition verdict is required: the zero-refusal models are fluent hallucinators that always produce a confident wrong-person biography, which the verdict scores as non-recognition, while embedding similarity—flat across judge-score buckets—would credit exactly those bios as recognized (Appendix L).

Orderings survive wording, vintage, and judge; levels do not. Re-probing under three paraphrased templates leaves the entity ordering and every cohort-level finding intact while shifting absolute levels, so NameRank values are template-conditional (Appendix M). A training-cutoff natural experiment on two contemporaneously-published author cohorts shows the silent zone is intrinsic rather than corpus timing—but also reveals a $\sim$0.13/yr cross-vintage drift, so longitudinal claims must be made on within-run *relative* gaps (Appendix N); the thresholded behavior itself is what the transformer fact-storage literature predicts [Geva et al., 2021, Petroni et al., 2019, Kandpal et al., 2023, Zhang et al., 2025]. And a three-judge re-grade (Gemini, Claude, GPT-5) preserves every finding, with only a small own-family scoring lift that shifts per-vendor tables (Appendix O).

Floors are calibrated and confounds rejected. Thirty fictional entities establish the noise floor: near-zero for dense golds, elevated for boilerplate golds—and floor-adjusting *widens* the credential gap (Appendix P). Dedicated ablations reject gold-answer length, context leakage, and Wikipedia presence or attention as alternative explanations: NameRank is not a has-a-Wikipedia-page flag—the binary flag explains $\sim$2% of long-tail variance against $\sim$40% for h-index—and it is not pageview rank, which is undefined for 71% of entities and explains $R^2 = 0.06$ where defined (Appendix Q).

5 Discussion

5.1 What We Have Built and What We Have Not

NameRank is a continuous cross-domain instrument with empirical threshold cleanliness, validated external correlates (h-index, institution, the event attention ledger), and mechanism-level structure. It is suitable for cross-domain placement on a common scale, for tracking name propagation into training corpora over time, and for measuring the recognition channel that bibliometrics miss. It is not a quality metric (it measures what models absorbed, not what is worth absorbing), not an attribution system (the name, not the deed), and not predictive of anything the corpus has not yet recorded.

5.2 The Credential Treadmill Thesis in Context

The credential treadmill does not contradict the olympiad-outcome literature, which finds medals causally raise subsequent *academic productivity* [Agarwal and Gaule, 2020, Lubinski et al., 2020]; our dependent variable is different. When it is name-propagation into LLM corpora rather than paper count, the credential per se is a near-zero predictor—continued production of named, indexable artifacts is what propagates. The credential still signals ability and motivation; only its role as a name-propagation channel has shrunk. In the language of signaling theory [Spence, 1973], the LLM-corpus channel transmits not the credential itself but the named, indexable production that may follow it.

5.3 What Clock Does NameRank Read? Lifetime vs. Current-Term Recognition

A recognition score could measure two different quantities: *cumulative* presence—everything a name has deposited in the corpus over a career—or *current-term* salience, what the world is paying attention to now. Three findings established independently above jointly place NameRank in the first category. Within IMO gold medalists, medal year is uncorrelated with the score (Pearson -0.04 over 2005–2015; Appendix B): recognition neither fades for older medalists nor spikes for recent ones. MSRA fellowship cohorts show the complementary shape—five-year-cohort means rise from 0.312 (2005–2009) to 0.382 (2015–2019) and dip only for the 2020–2024 fellows—exactly what an integral over post-credential production looks like when the newest cohorts have not yet had time to produce (Appendix B). And on news events, where attention is directly recorded, the score loads on *peak* salience rather than persistence (Section 3.6): what matters is how many name-bearing documents were minted, not how long the audience stayed. Read together: NameRank is a cumulative ledger of name-bearing text, snapshotted through the current model fleet—a lifetime measure weighted by documentation density, not a barometer of current attention. Current-term change is accessible only longitudinally, as within-run relative gaps across successive fleets (Section 5.9).

5.4 The Artifact $>$ Creator Inversion and the Anonymity-of-Tooling Problem

For independent creators of well-named single artifacts the tool out-ranks its maker (Section 3.4), and the context-injection experiment is consistent with this being a causal effect (modulo a coverage-echo confound we bound but do not fully remove, Section 3.4.4). The mechanism is attribution-direction asymmetry: creator-bios feature their creations, but artifact-discussion text (OSS docs, tutorials, papers) names the artifact far more often than the creator. The actionable consequence is uncomfortable but concrete: for propagating a name through the LLM channel, building a clearly-named tool beats writing many papers—yet the lift accrues to the tool's name first and to the creator only through it, so a tool buried under generic corporate branding yields no personal NameRank.

5.5 What Raises Recognition—and Whether Going Viral Helps

Read prescriptively, the findings converge on a single lever: recognition accrues to *distinctively-named, indexable production accumulated over a career*. Volume of name-bearing works dominates citation weight (h-index over raw citations, Section 3.5); one clearly-named artifact outperforms many papers (Section 3.4); and only the marquee credential tier propagates on its own, mid-career honors do not (Section 3.3). Because the ledger is cumulative (Section 5.3), the effective strategy is long-horizon and structural rather than about any single achievement: keep depositing documents that carry a distinctive, low-collision name into English-indexed text.

This also bounds what public attention buys. Going viral helps only insofar as the spike is *captured as durable, name-bearing documents*: on news events the score loads on peak salience over persistence and enters through the accuracy channel, not raw coverage (Section 3.6), so a burst that mints many correct name-attached records converts while ephemeral engagement that leaves none does not—and even captured attention decays (the -0.049 vintage echo). The conversion is category-dependent: within celebrity cohorts pageviews track NameRank strongly, but for technical creators attention *flow* and corpus *stock* decouple and can invert

(Appendix Q; a 6.4K-pageview library outscores a million-pageview model). Attention is thus a means, valuable only when it deposits distinctively-named text into the corpus—never an end in itself.

5.6 The Corpus-Density Gradient and Its Limits

Section 3.5.2 documents a twofold Stanford–Tsinghua recognition gap that recurs at country scale, with the well-sampled USA-over-China-over-India ordering as its firm backbone and the small-country leaders as suggestive only. The mechanism is English-corpus density; the consequence is structural inequality in name propagation per unit of research output.

Three caveats: the bottom-of-ladder institution samples are small ($n = 4$); uncorrected researcher mobility inflates the US average and depresses China's; and the gradient is downstream of the indexing and crawling decisions of training-data pipelines, so it could in principle be reduced by changing them. Within the current corpus regime, it is a substantial structural effect.

5.7 Inherited Demographic Biases

NameRank inherits the demographic biases of its judge and probed models, and measures them faithfully rather than correcting them. Li [2026] documented a $\sim 2\times$ recognition attenuation for common East-Asian surnames; we find a second, smaller effect on first-name-inferred gender: within the researcher cohorts—where institution matching and h-index-decile adjustment are available and survive—man-coded names outscore woman-coded names by roughly 0.03–0.05 (Appendix R). We restrict the claim to that controlled comparison: the much larger raw gaps in cross-profession cohorts reflect real fame differences among the sampled individuals, not a NameRank artifact. As with the surname effect, the gender gap is an inherited property of the corpus-and-judge stack that NameRank reports, not a calibration target; downstream uses must account for it.

5.8 Limitations

Appendix S tabulates every caveat with its direction and mitigation. The material ones: absolute levels are wording- and gold-conditional while rankings are robust; the full run uses a single judge with a small own-family lift; only within-run *relative* gaps are comparable across model fleets ($\sim$0.13/yr vintage drift); and the context-injection lift is an upper bound on named-artifact amplification.

5.9 Registered Predictions for Future Re-Runs

NameRank is a snapshot, so a re-run each release cycle yields a longitudinal trajectory. Because of the $\sim$0.13/yr vintage drift, we state predictions as within-run *relative* gaps (so the drift cancels) and register four falsifiable ones for the 2027 fleet: (i) the IMO-gold-vs-OpenAlex gap widens from -0.102 to ≤ -0.13 within 12 months (named-artifact-attached careers accumulate text, credential-only cohorts do not); (ii) the median artifact-minus-creator delta on independent-creator pairs grows from $+0.09$ to $\geq +0.14$ within 18 months; (iii) the Stanford-vs-Tsinghua gap narrows by $\geq 25\%$ but does not close within 24 months as Chinese-language training data enters frontier models; (iv) on a re-run of the news-event cohort (Section 3.6)—the cleanest of the four, since its exposure side is frozen in the pageview archive—the 2023-vs-

2021 vintage gap (-0.049 at matched attention) shrinks by at least half within 18 months as post-event derivative text accumulates.

6 Related Work

6.1 Relation to the Predecessor Instrument

The direct predecessor of this work is the recognition analysis in the IKP benchmark [Li, 2026, §5.7], which established the findings this paper builds on: on 345 CS-researcher probes, log(citations) explains only $\sim$35% of recognition variance, with the remainder attributed in a controlled audit to *named-artifact amplification*, *name uniqueness* (common East Asian surnames attenuate recognition by $\sim 2\times$), and *subfield-ecosystem density*; and on 557 Wikidata-grounded probes, domain-specific recognition multipliers trace to the structure of web discourse around each fact type rather than to entity prominence. IKP nonetheless treated recognition as a benchmark byproduct. NameRank differs in each dimension that matters for an impact metric: a $[0, 1]$ recognition rate over a 36-model panel (vs. a six-landmark tier ladder calibrated for parameter estimation); thousands of entities across dozens of cohorts spanning industry, OSS, and mid-tier figures (vs. 345 researchers and 557 Wikidata probes); open-ended probes scored by a per-model recognition verdict (vs. closed-factoid binary verdicts); h-index rather than raw citations as the validated bibliometric correlate; and cross-model variance retained as signal rather than absorbed into a tier label.

6.2 Scientometric Measures of Researcher Impact

The classical instruments for researcher impact—citation count, h-index [Hirsch, 2005], journal impact factor—measure intellectual reach within the academic publication system, and are increasingly understood as failing under pressures relevant here: authorship-list inflation and uneven attribution [Larivière et al., 2019], the *hyperauthorship* problem of pricing contributions to thousand-author papers [Cronin, 2001] (a GPT-4 co-author's citation count is mechanically large but says nothing about name-recognition), prestige-hierarchy biases spanning an order of magnitude across institutional tiers [Wapman et al., 2022], and within-researcher variance that motivates decompositions like the Q parameter [Sinatra et al., 2016].

None of these instruments cross the academic boundary cleanly. Citation count does not measure Sam Altman, Aravind Srinivas, or a Walmart CTO; h-index does not measure the creator of nanoGPT. Bespoke domain metrics (Google PageRank for websites, GitHub stars for OSS, Spotify monthly listeners for musicians, IMDb scores for film) do not interconvert. Collective-attention metrics do—search volume, Google Trends, and Wikipedia pageviews place any entity on one axis, and pageviews in particular are an established salience proxy [Mestyán et al., 2013, Ripberger, 2011]—but they count queries and article traffic rather than verified knowledge, and they are empirically near-orthogonal to NameRank: undefined off Wikipedia, weakly predictive where defined, and negatively correlated within technical cohorts (Appendix Q). We use attention records as calibration input on the event cohort (Section 3.6), not as the measurand: NameRank reads the *outcome* of the training pipeline—what deployed models can correctly state—rather than the exposure that feeds it. The cross-domain measurement of eminence has a long prior tradition in *historiometry* [Simonton, 1999], which quantifies the reputational standing of eminent individuals from archival traces; NameRank can be read as a historiometric instrument whose archive is the frontier-model training corpus rather than biographical dictionaries and citation indices. For impact assessment that spans academia, industry, OSS, and the long tail of solo operators, a cross-domain instrument is needed; NameRank is the first we are aware of that reads this archive directly.

6.3 LLM-as-Judge for Quality and Impact

Thelwall [2024, 2025b,a] use ChatGPT to score paper *quality* against the UK REF rubric; Kousha and Thelwall [2025] extend to societal influence; Liu et al. [2026] forecast citation counts. These differ from NameRank in two respects: they ask the LLM to *judge* quality, where we ask what the model has *absorbed*; and they average within a single model across iterations, where we average across the full 36-model panel, treating cross-model variance as signal.

A separate line asks whether models *know what they know*: self-assessed confidence is well calibrated on aggregate question answering [Kadavath et al., 2022] but degrades at the boundary of a model's own competence [Yin et al., 2023], and entity popularity modulates factual reliability [Mallen et al., 2023]. Section 3.8 adds an entity-level counterpart: pairwise self-reported recognition, aggregated with the Bradley–Terry machinery familiar from Chatbot Arena [Chiang et al., 2024], tracks the corpus salience shared across vendors rather than the reporting model's own verified knowledge.

The known LLM bias most relevant to our setting is the *Matthew effect* [Merton, 1968] as it propagates through LLM training corpora: frontier models recall already-highly-cited work disproportionately, inflating the rich-get-richer dynamic familiar from the bibliometrics literature. We inherit this bias and treat it as a known property of the measurand—NameRank measures *what frontier models have absorbed*, which is itself shaped by the Matthew dynamic. The bias is part of what we measure, not a flaw to be corrected.

6.4 Credentials and Long-Run Outcomes

A longitudinal literature finds that early mathematical talent and olympiad medals predict—and causally raise—later *academic* productivity: the 50-year SMPY study [Lubinski et al., 2020], a medal-cutoff natural experiment [Agarwal and Gaule, 2020], and work documenting substantial within-elite heterogeneity [Güllich et al., 2025]. Our credential-treadmill finding does not contradict this literature; it changes the dependent variable from academic production to LLM-mediated name recognition and finds that the credential per se carries little of it (Sections 3.2, 5.2).

7 Conclusion

NameRank operationalizes the 65% recognition-variance residual that Li [2026] characterized but did not measure: a cross-domain, hallucination-resistant recognition rate over a 36-model panel. Its findings each subvert a default expectation—olympiad credentials no longer propagate names, named tools out-rank their makers, the structure of citation production beats its volume, on news events, where world attention is directly recorded, peak salience rather than persistence drives propagation, and asking a model to rank its own recognition returns the shared corpus prior rather than its own knowledge state—and each is mechanistically grounded and survives a battery of robustness checks. The broader claim is that human curiosity about people and artifacts is now re-mediated by a few frontier models, and whether a name has propagated into their training corpora is itself a measurable quantity. Three uses follow: a cross-domain impact metric (a Walmart CTO at 0.52 is directly comparable to a Stanford professor at 0.54), a longitudinal tracker of how discourse reaches training pipelines, and a normative indicator for career strategy (build and name your own tool). All probes, gold answers, responses, and code are released at `https://github.com/19PINE-AI/namerank`; companion site `https://01.me/research/namerank`.

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The appendices are organized in three blocks. Appendix A completes the instrument specification of Section 2. Appendices B–K give the full tables and case detail behind each finding of Section 3, in the order the findings appear. Appendices L–R give the evidence behind the validation battery of Section 4, one appendix per threat, followed by the limitations summary (Appendix S).

A Probe, Panel, and Cohort Specification

Why disambiguation does not contaminate scoring. The probe context names role, affiliation, and field—never specific contributions, papers, dates, or named artifacts. A model that recognizes the entity produces specific facts, which the judge scores against the gold; a model that does not produces generic statements about the role, which the judge assigns low coverage because the rubric credits only *specific* named facts (Section 2.4). The sharpest empirical check comes from the GPT-5 system-card author cohort, whose disambiguating context is identical across all 80 entities: 71% of probes nonetheless return "unknown" rather than a generic in-context biography. A dedicated context-ablation experiment (Appendix Q) confirms the same conclusion.

Table 2. The 37-model panel, by vendor class. Thinking-mode variants are used wherever available.

Western (n=23)	**Chinese (n=14)**
GPT-5.3, GPT-5.4, GPT-5.5-think, GPT-OSS-20b-think	DeepSeek-V3.2-think, V4-flash-think, V4-pro-think
Claude Opus 4.6-think, Sonnet 4.6-think	Qwen3-8b-think, 32b-think, 235b-a22b-think, 3.5-397b-a17b-think
Gemini 2.5-pro-think, 3-flash-think, 3.1-pro	GLM-4-32b, 4.7-think, 5.1-think
Llama-3.1-8b, 3.2-1b, 3.3-70b, 4-Maverick	Kimi-K2, K2.6-think
Mistral-Large, Medium 3.1, Small 24b, Ministral 3b	Ernie-4.5-300b-a47b
Gemma-3-4b, 3-12b, 4-31b	MiniMax-M2.7-think
Grok-4, Grok-4.20-think, Phi-4	

Table 3. Entity cohorts by top-level group. Full per-cohort breakdown in the released cohort specification.

Group	n	Representative cohorts
Credentialed individuals	801	IMO/IOI/CMO/NOI/CPhO medalists, Putnam, ICPC, Rhodes, MSRA Fellowship
Working academics	1,628	CS faculty; OpenAlex researchers with 500–30,000 citations
Industry figures	248	founders, VCs, YC companies, AI-policy officials
Named artifacts	1,332	foundation models, OSS projects, papers, benchmarks, named methods
Mid-tier cross-profession	1,461	writers, athletes, actors, chefs, journalists, podcasts, books
Diagnostic reference set	37	hand-curated public figures, engineers, and artifacts
Special cohorts	212	GPT-5 system-card authors, DeepSeek-V3 authors, conferences, awards

B Credential Ladder Detail

Table 4 gives the full credential ladder summarized in Figure 3.

Table 4. Credential ladder. The OpenAlex long-tail researcher cohort ($n = 771$, citations 500–30,000) supplies the working-researcher baseline. The two rows marked † are large-team co-authorship cohorts shown for context, not credentials, and are excluded from the "7 of 9" count.

Credential	n	Mean NameRank	vs. baseline
ICPC World Finalist gold	18	0.610	+0.146
Putnam top-25 fellow	35	0.608	+0.144
OpenAlex long-tail researchers (baseline)	**771**	**0.464**	—
NOI China gold	29	0.368	−0.096
IMO gold (2005–2015)	**197**	**0.362**	**−0.102**
MSRA PhD Fellowship	237	0.343	−0.121
IOI gold	68	0.341	−0.123
Rhodes Scholarship	36	0.315	−0.149
CMO China gold	131	0.302	−0.162
CPhO China first prize	50	0.182	−0.282
DeepSeek-V3 paper author†	69	0.178	−0.286
GPT-5 system-card author†	80	0.042	−0.422

Career-track split. Within IMO gold medalists, US-tracked academic/industry careers—the subset most comparable to the citation-floored OpenAlex baseline—score ~0.50; non-Anglophone-tracked golds score ~0.30. The medal-and-name pair propagates only where the post-medal career produces English text.

Temporal patterns. Within MSRA fellows, five-year-cohort means rise from 0.312 (2005–2009) to 0.337 (2010–2014) to 0.382 (2015–2019) and dip to 0.343 (2020–2024), consistent with recent fellows having had less time for post-fellowship production to propagate. Within IMO 2005–2015, Pearson(year, NameRank) = −0.042, indistinguishable from zero: the credential signal neither strengthens nor decays within the ten-year window.

C Mid/Late-Career Awards as Recognition Predictors

This appendix gives the construction and full tables behind Section 3.3 and Figure 4.

Cohorts. Laureate rosters were pulled from Wikidata (award-received statements, with the award-date qualifier supplying the year): Turing Award (77 human laureates with an English label), Fields Medal (64), Nobel Prize in Physics restricted to 2000–2023 (64), Gödel Prize (80), ACM Prize in Computing (18), and, sampled to 60 each from their 2000–2023 Wikidata-listed recipients, the MacArthur Fellowship and ACM Fellow honors; the Sloan Research Fellowship (49 listed for 2000–2023) is the early-career anchor. The five prize cohorts (Turing, Fields, Nobel, Gödel, ACM Prize) are complete rosters by construction—every laureate has a Wikidata item. The three fellowship/honor cohorts (MacArthur, ACM Fellow, Sloan) are sampled from Wikidata's coverage, which omits lower-profile recipients, so their means are *upper bounds* and are read as such.

Panel and in-run controls. The run uses the 36-model 2026 panel of the news-event cohort (Appendix J: 28 main-run survivors + 8 IKP-v2 replacements). Because absolute levels shift across fleets (~0.13/yr drift, Appendix N), the ladder is anchored by controls re-probed on this same panel with contexts and gold answers copied verbatim from the main run: 60 OpenAlex long-tail researchers (the working-researcher baseline) and 40 IMO golds. All comparisons in Section 3.3 are within this run.

Golds and contexts. Contexts follow the main run's credential convention—award name and year only, never contributions (e.g., *"a mathematician who received the Fields Medal in 2014"*). Gold answers are the laureate's en-Wikipedia lead section trimmed to $\sim$200 words where available (372 of 472 laureates), and otherwise the boilerplate recipe used for the MSRA/IMO golds (award fact plus two-to-three sentences describing the award), so gold density mirrors the credential cohorts. The per-entity gold source is recorded and used in the robustness column below.

Table 5. The award ladder. Mean NameRank over the 36-model panel with $\pm$1 SEM; *Wiki-only* restricts to members carrying a dense Wikipedia-intro gold (the conservative subset, since dense golds cap coverage); *vs. base* is relative to the in-run OpenAlex baseline (0.494); *yr. corr* is Pearson(award year, NameRank) within the cohort. The three marquee late-career awards clear the baseline by $+0.15$ to $+0.20$ and show a negative year correlation (older laureates score higher); the mid-career honors and the IMO early credential sit at or below it.

Award	Stage	n	Mean	Wiki-only	vs. base	yr. corr
Nobel Prize in Physics (2000–23)	late	64	0.695	0.696	+0.201	−0.124
Turing Award	late	77	0.674	0.652	+0.180	−0.175
Fields Medal	late	64	0.640	0.657	+0.146	−0.218
ACM Prize in Computing	mid	18	0.563	0.509	+0.070	−0.215
Sloan Research Fellowship‡	early	49	0.519	0.384	+0.026	+0.253
OpenAlex researchers (baseline)	—	60	0.494	—	—	—
ACM Fellow‡	mid	60	0.473	0.400	−0.021	+0.368
Gödel Prize	mid	80	0.467	0.446	−0.027	−0.154
MacArthur Fellowship‡	mid	60	0.443	0.452	−0.051	+0.013
IMO gold (in-run control)	early cred.	40	0.386	—	−0.108	—

‡ Wikidata-coverage sample (omits lower-profile recipients); cohort mean is an upper bound.

The lead is not a gold-density artifact. Denser golds cap the coverage a 150-word response can achieve, so the marquee laureates—who carry rich Wikipedia intros—are if anything understated. The *Wiki-only* column confirms it: restricted to Wikipedia-gold members, Nobel/Turing/Fields remain at 0.696/0.652/0.657, far above baseline. Where boilerplate golds inflate a cohort (they are partly satisfiable from the context, the synthetic-null effect of Appendix P), it works *against* the finding: the mid/early cohorts' boilerplate members lift their means, and the marquee awards still lead.

The award adds signal beyond bibliometrics. On the 140 ladder members with a confident OpenAlex author match (name-similarity ≥ 0.85; the 80 marquee laureates plus the 60 OpenAlex controls), $\log_{10}(h\text{-index})$ alone explains $R^2 = 0.05$ of NameRank—the pool is compressed at high h, so bibliometric spread is small—while a marquee-award indicator explains $R^2 = 0.23$, and in joint regression the award adds $R^2 = 0.19$ beyond h-index, with a coefficient of $+0.158$ at matched $\log_{10}(h\text{-index})$. The award marks recognition the bibliometrics miss. Caveat: the OpenAlex match for a famous laureate probed by common name is imperfect (the top search hit may carry a truncated profile), so we read the regression as directional support for the ladder, not a calibrated effect size.

The lifetime signal. Within each marquee award, standardizing NameRank against the award's own mean and pooling isolates time since the award from each award's level and year-span. Binned by years since the award, mean standardized NameRank runs -0.51 (0–10 yr, $n = 41$), +0.10 (10–20, $n = 46$), +0.16 (20–30, $n = 44$), +0.25 (30–40, $n = 20$), +0.09 (40+, $n = 54$); the pooled Pearson correlation is +0.15. Recognition keeps accumulating for roughly three decades after even a Nobel, Turing, or Fields award, confirming the cumulative-ledger reading of Section 5.3. The positive within-cohort year correlations for ACM Fellow (+0.37) and Sloan (+0.25) run the other way but are sampling artifacts: for these Wikidata-coverage cohorts, the more recent recipients who already have an article are the more prominent ones.

Two utility websites (Section 3.7). Probed on the same panel: Hackergame 0.459, icourse.club 0.211, tuixue.online 0.182 (this panel), against nanoGPT 0.743 as the named-artifact anchor. Hackergame's lead over the two course/visa sites tracks its uniquely-named, GitHub-documented English-readable write-ups; the others spread functionally under anonymous-by-convention attribution.

D Medal Tiers: The Complete NOI 2009 Roster

The credential ladder above measures *gold* medalists only, so it cannot say whether the medal's grade carries signal. To test this we probed the complete medal roster of a single contest — NOI 2009: every gold, silver, and bronze medalist (103 in all) — on the 36-model 2026 panel of the news-event cohort (Appendix J). The roster comes from the official CCF medal list, cross-validated line-by-line against the community OIerDb database (identical names, schools, and scores); the released files include each medalist's contest score and rank. Probe names follow the main run's romanization conventions and carry the Chinese characters, which — together with the year-and-school context — disambiguate medalists whose Pinyin collides with common names.

Measurement protocol. This roster is scored under the recognition judge used throughout the paper (Appendix L). Each medalist's gold answer is an individual, web-grounded profile of who the person actually is — their real career and named contributions, retrieved by a search-augmented model and length-normalized to ∼55 words. The judge is shown the probe context and returns a binary *recognized* verdict: it credits recognition only when the response states a specific, non-guessable fact about the person that is true either per the gold or per the judge's own knowledge, and *not* derivable from the disambiguating context, so a response that merely restates "a 2009 NOI gold medalist from [school]" is not recognition. Because the entities are competition participants—for whom a judge has no reliable per-individual memory—the anti-confabulation provision applies: fine-grained minutiae asserted via the judge's own knowledge (other-year participations, exact scores) are not credited, only gold-present facts or major documented achievements. The per-entity score is the fraction of the 36-model panel that recognizes the person.

Most of the population sits near the floor, and that is the finding. The typical NOI medalist is recognized by only a handful of the 36 models: the gold-tier median is 0.056 (about two models), the silver median 0.028, the bronze median 0 (Table 6, Figure 7). Even in the gold tier, where 18 of 20 are recognized by at least one model, the mass is a thin near-floor band with a short recognized tail—a frontier panel stores only a small, documented slice of even this pre-selected population. This is Property 2 of the metric operating at full strength: against a synthetic-null floor of 0.000, the gold mean of 0.104 is a real above-floor signal, and the silver/bronze means (0.026, 0.019) sit just above it.

The gold/non-gold cliff survives; silver and bronze do not separate. The gold tier stands

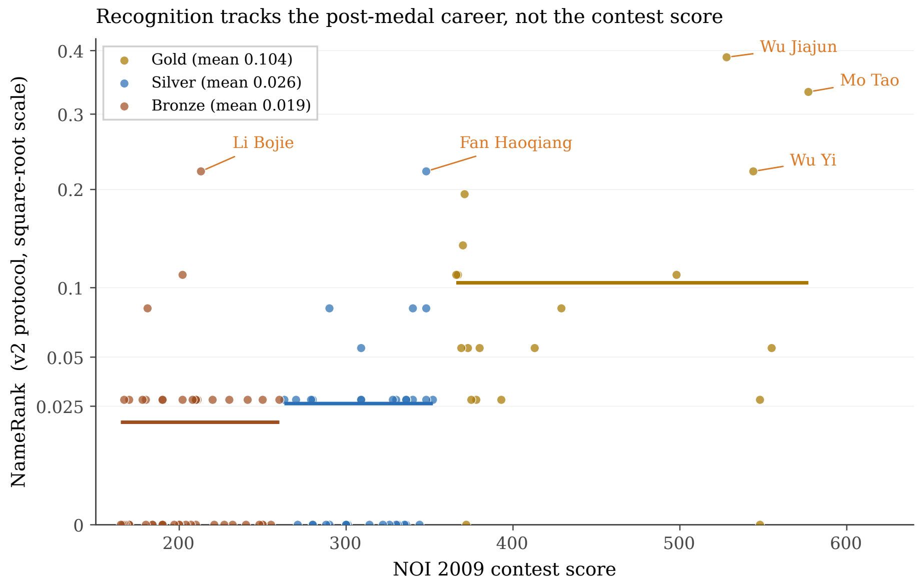

Figure 7. NameRank against NOI 2009 contest score for the complete medal roster (square-root y-axis, so the dense near-zero band and the recognized tail share one panel), with within-tier means as horizontal lines. Most medalists are silent ($\approx$0); only the five recognized medalists above 0.2 are labelled. Several golds scoring $\geq$500 stay near the floor unlabelled, while two much lower-scoring medalists (Fan Haoqiang, Li Bojie) are recognized—recognition tracks the post-medal career, not the contest score.

Table 6. NOI 2009 medal tiers under the recognition judge. The score is the panel recognition rate—the fraction of the 36 models that demonstrate genuine recognition (Appendix L); "silent" counts medalists no model recognizes, "recognized" those at least one model does. The protocol-matched synthetic-null floor is 0.000 (three fictional NOI medalists, same recipe), so the gold-tier mean is entirely above-floor signal. Kruskal–Wallis across tiers $H = 25.9$ ($p < 10^{-5}$); gold vs. silver Cliff's $\delta = +0.62$; silver vs. bronze indistinguishable ($p = 0.27$, $\delta = +0.13$).

Tier	n	Mean	Median	Silent (0 models)	Recognized ($\geq$ 1)
Gold	20	0.104	0.056	2/20	18/20
Silver	34	0.026	0.028	16/34	18/34
Bronze	49	0.019	0.000	28/49	21/49

clear of the lower tiers (Cliff's $\delta = +0.62$ vs. silver, Kruskal–Wallis $p < 10^{-5}$), while silver and bronze are statistically indistinguishable ($\delta = +0.13$, $p = 0.27$). The $\sim$20-person national gold roster is the documented one—admissions coverage, competitive-programming lore, team-selection reporting—and it carries by far the most recognition mass (mean 0.104 against 0.026 and 0.019). Contest score *within* a tier does not predict recognition (Figure 7): the effect is the roster cut, not the marginal point.

The recognized names are career-driven — the treadmill again. The medalists who rise above the near-floor band are exactly those whose *post-medal* careers produced named, indexed work: the most-recognized medalist overall is a gold who is now a Stanford computer-vision professor (Wu Jiajun, 0.39); the most-recognized silver matches the gold-tier top on a prominent vision-research career (Fan Haoqiang, 0.22, later an IOI gold and a Megvii research lead); and the most-recognized bronze ranks near the bottom of the contest's score column yet ties the top silver (Li Bojie, 0.22, the author of this paper). The medal grade sets the odds of entering a documented cohort; the recognition itself tracks what the person later built.

One gold per person makes the score stable across framings. A by-product validates the corrected golds. The bronze medalist above also appears in the main run under two other probe framings (a fellowship recipient; an AI-infrastructure researcher). Graded against the old cohort-boilerplate golds, the same person scored 0.35, 0.41, and 0.09 — a threefold swing driven entirely by how much each gold demanded. Graded against a single web-grounded profile with the recognition judge, the three framings converge to within 0.04 of one another: the same person now carries essentially one score regardless of how the prompt introduces them, which is the property a recognition measure should have.

The noise floor is calibrated with protocol-matched synthetic-null entities—three fictional NOI 2009 medalists built on the same gold recipe and scored by the same judge—and comes out at 0.000: the anti-confabulation provision leaves fictional competitors with no recognized facts at all. The gold-tier mean of 0.104 is therefore an entirely above-floor signal, and the silver (0.026) and bronze (0.019) means are small but non-zero above a hard floor.

E Artifact–Creator Pairs: Inversion, Injection, and Asymmetry Detail

This appendix gives the full tables behind Figure 5 and the Tianshou case study (Section 3.4).

E.1 The inversion and injection tables

Table 7. The artifact-over-creator inversion. NR_c and NR_a are creator and artifact NameRank; $C \to A$ is the fraction of responses about the creator that mention the artifact, $A \to C$ the reverse. Boldface: artifact exceeds creator.

Creator	NR_c	Artifact	NR_a	$\Delta(a-c)$	$C \to A$	$A \to C$
Simon Willison	0.594	**Datasette**	**0.899**	**+0.305**	0.54	0.89
Dario Amodei	0.584	**Anthropic**	**0.811**	**+0.228**	1.00	0.41
Andrej Karpathy	0.614	**nanoGPT**	**0.707**	**+0.093**	0.05	0.76
Jiayi Weng	0.331	**Tianshou**	**0.420**	**+0.089**	0.25	0.00
Tri Dao	0.532	**FlashAttention**	**0.607**	**+0.075**	0.74	0.54
Lilian Weng	0.590	**lilianweng.github.io**	**0.614**	**+0.024**	0.97	0.97
Harrison Chase	0.479	**LangChain**	**0.502**	**+0.024**	1.00	0.22
Aman Sanger	0.289	**Cursor**	**0.305**	**+0.016**	1.00	0.00
Demis Hassabis	0.664	Google DeepMind	0.490	−0.174	1.00	0.62
Mira Murati	0.442	Thinking Machines Lab	0.222	−0.220	0.97	1.00
Aravind Srinivas	0.663	Perplexity	0.193	−0.470	1.00	0.33

The injection confound in full. Because the injected clause names the artifact and the gold answer also names it, a model can raise its coverage score simply by echoing the injected token, without any additional stored knowledge—the same scoring-leakage channel otherwise closed by withholding contributions from the context (Section 2.2), deliberately opened here. Two features of the result limit this confound without eliminating it: the lift concentrates in low-baseline creators rather than spreading uniformly (pure echo would credit saturated leaders too, and they show none), and the tier gradient tracks recognition deficit (frontier-reasoning

Table 8. Context-injection experiment. Δ is the creator's NameRank change when the artifact name is added to the disambiguating context (configuration B − configuration A), with the paired t-statistic across the model panel. Bottom block: mean lift by model tier.

Creator	Artifact injected	Δ	paired t
Jiayi Weng	Tianshou	+**0.288**	5.8
Harrison Chase	LangChain	+**0.151**	3.3
Tri Dao	FlashAttention	+**0.094**	2.5
Simon Willison	Datasette	+**0.073**	2.8
Aman Sanger	Cursor	+0.042	1.8
Lilian Weng	lilianweng.github.io	+0.029	0.8
Demis Hassabis	Google DeepMind	+0.023	0.7
Andrej Karpathy	nanoGPT	+0.018	0.7
Mira Murati	Thinking Machines Lab	−0.014	−0.3
Aravind Srinivas	Perplexity	−0.022	−0.9
Dario Amodei	Anthropic	−0.047	−2.2
Mean over pairs		+0.058	
Lift by model tier	frontier-reasoning	+0.008	
	frontier-chat	+0.046	
	mid-weight	+0.106	
	small	+0.095	

+0.008, frontier-chat +0.046, mid-weight +0.106, small +0.095). The clean test—re-judging configuration B against an artifact-redacted gold that credits only facts *not* present in the injected clause—is left to future work; pending that, the measured lift is an upper bound.

E.2 The Tianshou case

Table 9. The Tianshou case: Jiayi Weng against eight OpenAI individual-contributor peers who share the affiliation but not a named artifact. Refusal rate is the fraction of the panel answering "unknown".

Entity	NameRank	Refusal rate
Jiayi Weng	**0.331**	24%
Tianshou (his artifact)	0.420	3%
Trevor Blackwell	0.431	8%
Vicki Cheung	0.300	24%
Pamela Vagata	0.184	32%
Adam Fry	0.084	51%
AJ Ostrow	0.078	60%
Andrea Vallone	0.076	57%
Alex Wei	0.039	51%
Aaditya Singh	0.027	68%
Peer mean	0.152	—
Jiayi Weng within peers	$+1.24\sigma$	—

For each of the 11 verified (creator, artifact) pairs, the full per-model breakdown of $C \to A$ and $A \to C$ (fraction of non-refusal model responses that name the counterpart) is released as a supplementary CSV.

Per-model mention table for Jiayi Weng / Tianshou. Of the 28 non-refusal responses to "tell me about Jiayi Weng," 7 name Tianshou: claude-opus-4.6-think, claude-sonnet-4.6-think, gemini-3-flash-think, gemini-3.1-pro, glm-5.1-think, gpt-5.5-think, kimi-k2.6-think. All other models that respond do not name Tianshou. The 7 mentioning models have judge scores ranging 0.70 to 0.80 (mean 0.71); the 21 non-mentioning responding models range 0.00 to 0.50 (mean 0.35). The score lift from artifact-mention is +0.36.

F External Validity Regressions

Full regression results for NameRank on external impact metrics, restricted to the OpenAlex long-tail researcher cohort ($n = 771$, citations 500–30,000, h-index from OpenAlex).

Predictor set	Full-sample R^2	10-fold CV R^2 (mean $\pm$ sd)
log(citations)	0.137	0.12 ± 0.07
log(h-index)	0.398	0.38 ± 0.09
log(citations) + log(h-index)	0.398	0.38 ± 0.09

Cross-validation is repeated 10-fold (20 repeats, 200 folds, fixed seed); the $\pm$ is the across-fold standard deviation. We use repeated CV rather than a single 70/30 split because at $n = 771$ a single split's held-out R^2 varies by $\sim$0.15 across seeds (one favorable split gave 0.50), which would overstate generalization. The marginal R^2 of log(citations) given log(h-index) is 0.000; the coefficient on log(citations) in joint regression is -0.002, indistinguishable from zero, and citations add no held-out signal.

Decile breakdown of NameRank by h-index (Table 10):

Table 10. Mean NameRank by h-index decile within OpenAlex long-tail researchers.

Decile	n	Mean h	Mean NameRank
D1	77	2.9	0.219
D2	77	8.1	0.329
D3	77	12.4	0.392
D4	77	16.7	0.448
D5	77	20.9	0.464
D6	77	25.8	0.524
D7	77	30.6	0.539
D8	77	36.9	0.574
D9	77	45.4	0.546
D10	77	60.4	0.601

The relationship is near-monotonic (a single dip at D9) and saturates above $h \geq 40$. The bottom decile ($h < 5$) shows NameRank floored at 0.22—the working-researcher floor, not zero.

G Bibliometric Predictor Decomposition

Six bibliometric predictors regressed on NameRank for the $n = 771$ OpenAlex long-tail cohort (Section 3.5), testing whether h-index dominates because it discounts per-paper name attribution. `fractional_citations` divides each paper's citations by its author count; `first_last_citations` counts only papers where the researcher is first or last author; `n_works` is the raw count of works; `mean_authors` is the mean author count per paper. All R^2 on $\log(1+x)$-transformed predictors.

Table 11. Sole-predictor R^2 on NameRank and marginal R^2 when added to h-index alone. Attribution-weighted measures (fractional, first/last) do not approach h-index, and add almost nothing on top of it; author count carries no signal. The count of distinct works (n_{works}) is the closest cousin of h-index.

Predictor	Sole R^2	ΔR^2 added to h-index
$\log(h\text{-index})$	**0.398**	—
$\log(n_{\text{works}})$	0.337	+0.0003
log(first/last-author citations)	0.268	+0.0268
log(fractional citations)	0.154	+0.0026
log(raw citations)	0.137	+0.0001
log(mean authors/paper)	0.0004	+0.0008

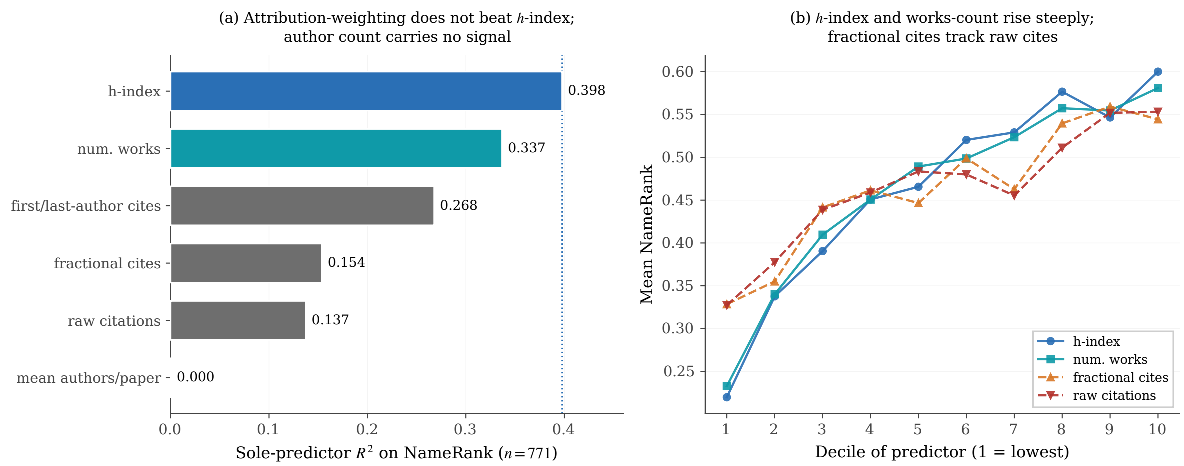

Figure 8. Bibliometric predictor decomposition ($n = 771$). **(a)** Sole-predictor R^2 on NameRank: if h-index dominated by discounting per-paper attribution, the attribution-weighted measures (fractional and first/last-author citations) would match it; instead they barely beat raw citations, and mean authors-per-paper carries no signal. The count of distinct works (n_{works}) is the closest cousin of h-index. **(b)** Mean NameRank by predictor decile: h-index and n_{works} rise steeply and near-monotonically; fractional citations track raw citations on a flat slope.

In a joint OLS over all six (total $R^2 = 0.428$), the standardized coefficients are dominated by $\log(h\text{-index})$ (0.579), with first/last-author citations (0.202) a distant second and raw citations *negative* (-0.143); fractional citations (0.109) and mean authors (0.060) are minor. The mechanism is name-recurrence across distinct works, not attribution-per-paper weighting: if the latter drove the result, fractional and first/last-author citations would match or exceed h-index, and they do not.

H Institution and Country Gradient Detail

Table 12 gives the institution-level view behind the Stanford–Tsinghua contrast; Table 13 the full country-level aggregation behind Figure 6.

Table 12. CS-faculty NameRank by institution (selected institutions with $n \geq 4$). Stanford-affiliated faculty score roughly twice Tsinghua-affiliated. The Chinese-institution samples are small ($n = 4$ each); the gradient they anchor recurs at country level (Figure 6), where the samples are larger.

Institution	n	Mean	Min	Max
Stanford University	19	**0.537**	0.14	0.62
University of Washington	26	0.529	0.13	0.68
Cornell University	33	0.484	0.16	0.67
Carnegie Mellon University	65	0.484	0.09	0.66
HKUST	5	0.462	0.24	0.58
Georgia Institute of Technology	6	0.462	0.23	0.65
Johns Hopkins University	4	0.453	0.19	0.67
TU Delft	5	0.445	0.30	0.58
Princeton University	25	0.443	0.18	0.61
University of Michigan	31	0.429	0.16	0.67
Universidade de Lisboa	5	0.427	0.26	0.52
Monash University	6	0.386	0.27	0.65
USTC	4	0.310	0.11	0.45
Peking University	4	0.290	0.22	0.34
Tsinghua University	4	**0.258**	0.15	0.35

The well-sampled backbone of the gradient is USA (mean 0.460, 95% CI $[0.443, 0.476]$) against China (0.311, $[0.271, 0.350]$) and India (0.292, $[0.234, 0.351]$): the intervals do not overlap. The apparent leaders (Israel, Singapore, Sweden) sit above the USA in point estimate with intervals that overlap it, and are read as suggestive only.

I Cross-Language and East–West Detail

Sub-run cohort selection. The Chinese-prompt sub-run covers 240 entities: the full diagnostic reference set (37), all NOI China gold medalists (29), all DeepSeek-V3 paper authors (69), 30 random samples each from the CMO, CPhO, and MSRA Fellowship cohorts, and 5 each from the writer, filmmaker, and politician cohorts as Western controls—$240 \times 37 = 8{,}880$ records.

English-prompt East–West extremes. On English prompts, the strongest Chinese-panel lifts over the Western panel are Chinese-content entities: Wei Dongyi (Chinese mathematics figure; Chinese-panel mean 0.80 vs. Western 0.53) and Zhang Fang (NOI gold; 0.45 vs. 0.20). The strongest Western-panel lifts are non-Chinese names on which Chinese models refuse: Saad Albawi (0.59 Western vs. 0.16 Chinese) and Qingzi Vera Liao (Chinese-name CS faculty at a Western institution; 0.50 vs. 0.15).

Per-model prompt-language deltas. The Chinese-prompt shift is universal rather than vendor-specific: models gaining on Chinese prompts include Western (claude-sonnet-4.6-think $+0.17$, gemini-3.1-pro $+0.12$, gemma-3-12b $+0.12$) and Chinese (deepseek-v4-pro-think $+0.14$, qwen3-235b $+0.10$) models, and models losing include both classes as well (llama-4-maverick -0.20, gpt-5.4 -0.12, glm-4-32b -0.05, kimi-k2 -0.05). The class averages are $+0.014$ (Western)

Table 13. CS-faculty NameRank by country of institutional affiliation, for countries with ≥ 3 sampled faculty. An *Other/Unknown* bucket ($n = 236$; CS-Rankings affiliations not matched by the keyword-prefix country lookup) is omitted. Only the USA ($n = 302$), China (30), UK (16), Australia and Hong Kong (11), and Brazil and India (10) reach $n \geq 10$; the remaining rows are point estimates with wide intervals and should not be finely ranked against one another.

Country	n	Mean	Median	frac ≥ 0.5
Israel	7	0.506	0.528	57%
Singapore	7	0.497	0.480	43%
Sweden	5	0.492	0.490	40%
Germany	9	0.490	0.488	44%
Switzerland	4	0.489	0.467	50%
Canada	9	0.478	0.463	22%
USA	302	0.460	0.488	46%
Netherlands	9	0.447	0.432	33%
UK	16	0.438	0.440	38%
Hong Kong	11	0.431	0.452	36%
Portugal	5	0.427	0.442	20%
Brazil	10	0.388	0.409	10%
Australia	11	0.341	0.291	18%
France	3	0.312	0.341	0%
China	30	**0.311**	0.285	7%
Spain	8	0.303	0.288	13%
India	10	0.292	0.279	0%

Table 14. Cross-language NameRank by cohort: $\Delta = \mathrm{NR}_{zh} - \mathrm{NR}_{en}$ on the 240-entity Chinese-prompt sub-run. Chinese prompts lift Chinese-named cohorts and slightly suppress English-coded ones; per-entity extremes in Appendix I.

Cohort	n	NR_{en}	NR_{zh}	Δ
DeepSeek-V3 authors	69	0.178	**0.247**	$+$**0.069**
Filmmakers (control)	5	0.765	0.802	$+0.037$
NOI China gold	29	0.368	0.392	$+0.025$
CMO China gold	30	0.301	0.315	$+0.013$
CPhO China first prize	30	0.184	0.194	$+0.010$
Politicians (control)	5	0.361	0.364	$+0.003$
Writers (control)	5	0.650	0.631	-0.020
MSRA PhD Fellowship	30	0.339	0.319	-0.021
Diagnostic reference set	37	0.452	0.422	$-$**0.030**

and $+0.028$ (Chinese): both lift slightly, and the dominant signal is entity-name match, not model origin.

Per-entity deltas. The full per-entity $\Delta(\mathrm{NR}_{zh} - \mathrm{NR}_{en})$ table is released as a supplementary CSV. Selected diagnostic entries (the largest suppressions beyond the table: Leng Junsheng -0.17, Du Zhuofan -0.15, Aleksander Madry -0.12):

Entity	NR_{en}	NR_{zh}	Δ	Cohort
Jiayi Weng	0.331	0.327	-0.004	reference set
Tianshou	0.420	0.399	-0.021	reference set
tuixue.online	0.250	0.262	$+0.012$	reference set
Andrej Karpathy	0.614	0.549	-0.066	reference set
Tri Dao	0.532	0.488	-0.044	reference set
FlashAttention	0.607	0.470	-0.137	reference set
Bojie Li	0.090	0.105	$+0.014$	reference set
Wang Haoyu (Wang Hao-yu)	0.260	0.430	$+0.170$	CMO China gold
Yao Xuanrong (Yao Xuan-rong)	0.250	0.420	$+0.160$	NOI China gold
Wang Zhongjian (Wang Zhong-jian)	0.260	0.420	$+0.160$	CMO China gold
Zijia Zhu	0.100	0.290	$+0.190$	DeepSeek-V3 author
Peng Zhang	0.160	0.330	$+0.170$	DeepSeek-V3 author

The pattern is consistent: Chinese-character names lift on Chinese prompts; English-coded artifacts lose on Chinese prompts; flat-cross-language entities have name attribution roughly equal in both training corpora.

J News-Event Cohort: Construction and Detail

This appendix documents the construction of the 258-event cohort of Section 3.6 and reports the full tables behind its findings. All construction scripts, inputs, and derived data are released under `experiments/t4_1_news_events/` in the repository.

J.1 Sampling frame and filters

Candidate events are harvested from the "Events" sections of the English Wikipedia year articles (*2021*, *2022*, *2023*) and of 32 country-year articles ("*2021 in India*", "*2022 in Nigeria*", ...) chosen to balance nine world regions (2,245 unique linked articles). Each candidate is classified by Gemini 3 Flash Preview (the study's judge family; temperature 0, structured output) from its title, short description, and first paragraph into: discrete event or not, start year and month, primary country, region, and category. An article qualifies as a *discrete event* if it describes a specific occurrence or tightly bounded series (an election, a storm, a trial, a tournament, an accident, a protest wave) that began at an identifiable date; people, organizations, products, laws, ongoing multi-year topics, and list/timeline articles are excluded (1,041 qualify). Post-classification filters require a standard-type article with an intro of ≥ 40 words and a first-year pageview record of $\geq 1{,}000$ total views over ≥ 200 recorded days; 655 events survive. Stratifying by $\log_{10}$(total views) band ($< 10^4$ through $\geq 10^7$) crossed with eight region groups, capped at 14 events per cell, yields the released cohort of 258.

J.2 Attention measures and cross-checks

For each event we retrieve the daily en-Wikipedia pageview series of its article (Wikimedia REST API, user traffic only) for the 365 days from the first day of the event's start month, and define **total attention** (summed daily views), **peak attention** (maximum daily views), and **effective duration** = total/peak in days, linked by the identity $\log(\text{total}) = \log(\text{peak}) + \log(\text{duration})$. Wikipedia pageviews are a standard collective-attention proxy [Mestyán et al., 2013, Ripberger, 2011]; we prefer them to Google Trends because the pageview record is absolute, archived, and reproducible, whereas the public Trends interface returns only normalized indices under heavy rate limits. Two caveats. First, the ledger is attention *to the event's English Wikipedia article*, so all matched-attention comparisons condition on English-audience interest; this is a feature for the region analysis (it holds English exposure fixed) but means globally quiet, locally enormous events enter as quiet. Second, attention can fragment across sibling articles (*Killing of Tyre Nichols* vs. *Tyre Nichols protests*), adding measurement noise that biases dose–response estimates toward zero. A GDELT news-coverage-volume cross-check [Leetaru and Schrodt, 2013] on the event's name phrase over the same window is reported under Full Results below and released with the data.

J.3 Probes and panel

Golds are the article's intro truncated to 100–200 words at a sentence boundary (the main run's Wikipedia gold recipe); disambiguating contexts are generated deterministically from the classification fields only—category phrase, location, month, year—never from article content, mirroring the main run's role/affiliation/field rule. Probe template, judge model, judge prompt, refusal detection, and scoring are unchanged from Section 2.

Panel. Nine of the 37 main-run models could not complete the event run (2026-07): eight lost all OpenRouter routing to provider delistings and deprecations (Ernie-4.5-300b, Mistral-Large-2411, Mistral-Medium-3.1, Ministral-3b, Kimi-K2, Grok-4, GLM-4-32b, Llama-3.2-1b), and Grok-4.20-think—initially recovered via the direct xAI API—lost access when that account exhausted its credits at 143 of 258 events, so its partial coverage is discarded (a model must cover the full cohort to enter the panel mean). The eight delisted models are replaced by the representative newly released models evaluated in the IKP revision [Li, 2026]: Claude Fable 5, GLM-5.2, Qwen3.7-Max, Kimi-K2.7-code, MiniMax-M3, Step-3.7-Flash, Nemotron-3-Ultra-550b, and Nemotron-3-Nano-30b (one representative per newly covered vendor family, plus a second NVIDIA entry to restore the small-model tier lost with Ministral-3b and Llama-3.2-1b; all run in thinking mode per the panel policy). Event NameRank is therefore a 36-model panel mean (28 survivors + 8 replacements) over 9,288 records. Neither change matters for comparability. Recomputing the full main run on the 28-model surviving sub-panel correlates with the released 37-model scores at $r = 0.993$ per entity (mean shift $+0.011$), so main-run numbers are quoted unadjusted alongside event scores, and cross-run placements use survivor-only event means. And the replacements reproduce the survivors' ordering: per-event means over the 8 new models correlate at $r = 0.905$ with means over the 28 survivors (level shift $+0.091$—the 2026-vintage models are uniformly more knowledgeable about 2021–2023 events), the entity-not-panel property of Appendix L on models released a year apart.

Cutoff coverage. Three panel models have training cutoffs inside the final months of the event window (Mistral-Small-24b, 2023-10; Llama-3.1-8b and Llama-3.3-70b, 2023-12); the eight replacement models are 2026 releases with cutoffs far past the window. Neither exclusion matters: dropping the 22 events that begin in October–December 2023 leaves the joint attention decomposition at peak $+0.045$ / duration -0.007; dropping the three models for all events gives $+0.038$ / -0.009 (vs. $+0.042$ / -0.008 headline).

J.4 Full results

Dose–response. Table 15 gives the decile ladder with the coverage/accuracy channel decomposition. Sole-predictor R^2 on NameRank: log total views 0.095 (repeated 10-fold CV 0.09 ± 0.11), log peak 0.119 (0.11 ± 0.12), log effective duration 0.033 (0.02 ± 0.08), log tail interest (mean daily views in days 300–365) 0.065. Gold length alone explains 0.006; adding it to log total views raises R^2 to 0.168 and the attention slope from $+0.042$ to $+0.063$ per decade (gold density suppresses).

Table 15. NameRank by decile of total first-year pageviews ($n = 258$). Attention converts refusal into accurate specifics: accuracy rises steeply, coverage gently.

Decile	n	Geo. mean views	NameRank	Refusal	(cov, acc)
D1	25	2,415	0.379	0.12	(0.46, 0.61)
D2	26	7,395	0.385	0.11	(0.47, 0.60)
D3	26	16,514	0.405	0.12	(0.46, 0.64)
D4	26	39,348	0.464	0.07	(0.54, 0.67)
D5	26	67,583	0.427	0.06	(0.49, 0.71)
D6	25	133,127	0.513	0.05	(0.57, 0.75)
D7	26	251,374	0.484	0.04	(0.55, 0.78)
D8	26	480,323	0.478	0.01	(0.53, 0.85)
D9	26	1,223,071	0.490	0.01	(0.54, 0.85)
D10	26	3,312,033	0.499	0.01	(0.54, 0.88)

Peak vs. duration. Joint regression of NameRank on the standardized log-components: peak $+0.042$ (HC1 se 0.008), duration -0.008 (se 0.009), joint $R^2 = 0.122$ (CV 0.11 ± 0.12); marginal R^2 of duration given peak $= 0.003$. With recurring-name and gold-length controls: peak $+0.070$ (se 0.012), duration $+0.005$ (se 0.010). Within peak terciles, moving from the shortest to the longest duration tercile moves NameRank by -0.07, $+0.01$, and -0.01 respectively—no duration channel at matched peak.

Recurring-series flag. An event is a series instance if substituting a nearby year (or adjacent ordinal/Roman edition marker) into its title yields another standard Wikipedia article; the detector sweeps $y \pm 1..6$ to cover 4–6-year election cycles. 106 of 258 events flag as recurring; at matched attention the penalty is -0.061 (se 0.015). The flag is conservative—renamed series (*2023 Men's FIH Hockey World Cup* vs. *2018 Men's Hockey World Cup*) escape detection and dilute the estimate.

Region and vintage. Table 16 reports raw and attention-adjusted region contrasts. Event-year effects at matched attention: 2022 vs. 2021: -0.009 (se 0.019); 2023 vs. 2021: -0.049 (se 0.018).

GDELT cross-check. Of the 258 name-phrase queries, the rate-limited GDELT DOC API returned data for 159; 118 had nonzero matched coverage. On those, log GDELT coverage volume correlates with log pageviews at $r = 0.33$—the two ledgers partially agree—but predicts NameRank at only $R^2 = 0.006$. We attribute the gap mostly to query noise: GDELT matches the event's literal name phrase, which misses paraphrase references and truncates awkwardly for long titles, and the returned subset is rate-limit-selected rather than random. We therefore rely on the pageview ledger and report GDELT as a weak consistency check on the attention record, not an independent predictor.

Table 16. Region groups: raw mean NameRank and the adjusted delta vs. the Anglo-America & Oceania baseline after controlling $\log_{10}$ total views (HC1 standard errors). The single Global-classified event (the Queen's Platinum Jubilee Beacons) is omitted.

Region group	n	Raw mean	Adj. Δ (se)
Anglo-America & Oceania	45	0.463	(baseline)
Eastern Europe & Russia	28	0.506	+0.038 (0.030)
Middle East & Africa	35	0.480	+0.009 (0.028)
Western Europe	44	0.453	−0.013 (0.025)
Latin America	32	0.452	−0.008 (0.026)
East Asia	31	0.425	−0.033 (0.031)
South & Southeast Asia	42	0.397	−0.060 (0.028)

K Self-Reported Recognition: Design and Detail

This appendix gives the design, estimation, and full results behind Section 3.8 (released as `experiments/t5_4_self_report/`).

K.1 Design

Entities. 430 total: 400 real entities sampled from the main run, stratified over six panel-NameRank bands ($[0, .05)$, $[.05, .15)$, $[.15, .30)$, $[.30, .50)$, $[.50, .70)$, $[.70, 1]$ with quotas 40/40/70/90/90/70 and at most 12 per cohort), plus the 30 fictional entities of Appendix P as traps.

Pairs. 3,350 unique pairs in five strata: 2,200 with both entities in the recognized region (panel NameRank ≥ 0.15; half constrained to close gaps $|\Delta| < 0.15$), 500 crossing the recognition boundary (≥ 0.30 vs. < 0.05), 250 with both below 0.15, and 400 trap pairs (each fictional entity against 12 real partners spread over the bands, plus 40 fictional–fictional pairs). A further 335 pairs (10%) are re-presented in swapped order as a position-bias probe, giving 3,685 presentations per model. Presentation order (A/B) is randomized once and shared across models.

Probe. Each pair is shown with the main-run disambiguation contexts, and the model is asked which entity it knows more about—"specific, verifiable facts such as works, roles, achievements, or dates"—answering exactly one of `A`, `B`, `EQUAL` (knows both to similar depth), or `NEITHER` (recognizes neither). The three self-report models are the main-run checkpoints GPT-5.5, Claude Opus 4.6, and Gemini 3.1 Pro at main-run settings, so each model's verdicts can be compared against its *own* released per-record behavioral scores. All 11,055 calls returned a parseable verdict.

Estimation. Each model's `A/B/EQUAL` verdicts on real–real pairs are fit with Davidson's tie-extended Bradley–Terry model [Bradley and Terry, 1952, Davidson, 1970]—the batch analogue of the Chatbot Arena aggregation [Chiang et al., 2024]—by ridge-regularized maximum likelihood; `NEITHER` verdicts are treated as abstentions. Confidence intervals are 95% normal-approximation pair-bootstrap (200 refits). Entity-level statistics use entities with at least five decided comparisons. Per-model labels for the binary analyses come from the model's own main-run record: *known* = score ≥ 0.15 and no refusal; *unknown* = refusal or score < 0.05.

Table 17. Self-reported recognition per model. ρ_{panel}: Spearman correlation of the Bradley–Terry self-ranking with panel NameRank over real entities (bootstrap 95% CI). Dir. acc.: fraction of decided verdicts agreeing with the panel ordering on both-known pairs with ΔNameRank ≥ 0.15. Boundary: known-vs-unknown pairs by the model's own labels—share picking the known side / the unknown side / abstaining. Trap: share of fictional-vs-known pairs where the fictional side was preferred (ties included). Reversal: verdict consistency under A/B swap.

Model	ρ_{panel}	Dir. acc.	Boundary (pick/err/abstain)	Trap	Reversal
GPT-5.5	0.61 [.59, .63]	0.82	0.74 / 0.12 / 0.14	1/232	0.88
Claude Opus 4.6	0.53 [.50, .56]	0.81	0.63 / 0.14 / 0.23	0/241	0.85
Gemini 3.1 Pro	0.57 [.55, .59]	0.81	0.72 / 0.09 / 0.16	0/211	0.90

K.2 Results

Partial reconstruction of the panel ordering. Table 17: the self-rankings correlate $\rho = 0.53$–0.61 with panel NameRank and agree with the panel ordering on 81% of decided both-known pairs at gaps ≥ 0.15. Position bias is negligible (reversal consistency 0.85–0.90; first-position win rate 0.46–0.49).

What self-report reads. Panel NameRank factors into a *prevalence* component (the share of the 37 models that know the entity, i.e. score ≥ 0.15) and a *conditional level* (mean score among the models that know it). The self-rankings load on prevalence ($\rho = 0.72$–0.76) and only weakly on conditional level ($\rho = 0.11$–0.19); the models' own behavioral scores do the reverse (conditional level $\rho = 0.60$–0.71, prevalence $\rho = 0.07$–0.15 among known entities). Consequently the self-ranking is nearly uncorrelated with the model's own scores among entities it knows ($\rho = +0.04$ / -0.06 / -0.09), while the three vendors' self-rankings agree pairwise at $\rho = 0.90$–0.91—each model agrees with the *other vendors' introspection* far better than with its *own behavior*. The conditional level is gold-conditional rather than fame-driven (specific dense golds are harder to cover than boilerplate golds; cf. the level-sensitivity in Appendices M and P), which is also why directional accuracy against raw own-score gaps lands below chance (0.34–0.40): raw per-model scores are not a cross-entity depth scale, and we do not use them as one.

Fame-incongruent boundary pairs. Restricting known-vs-unknown pairs to those where corpus fame contradicts the model's own state (the unknown entity has the higher panel NameRank by ≥ 0.05): abstention rises from 10–20% (congruent) to 35–64%, the famous-but-unknown side is claimed in only 0–9% of verdicts, and only GPT-5.5 still sides with its own knowledge on most such pairs (0.65, vs. 0.36 for both others; $n = 26/14/69$, so these splits are indicative). Self-report degrades into abstention, not fame-following fabrication.

Trap arm. Across 684 fictional-vs-known pairs the fictional entity was preferred once (GPT-5.5, 0.4% of its 232; ties included); `NEITHER` was answered on 95–100% of fictional–fictional pairs. Recognition claims are near-perfectly precise; the error budget is spent on under-claiming (abstention on 14–23% of genuine boundary pairs).

Caveats. The behavioral ground truth is one judged response per (entity, model), so response-level noise attenuates all within-model comparisons; the prevalence/conditional decomposition and the panel-side statistics are unaffected. `EQUAL` usage varies by vendor (3% of GPT-5.5 verdicts, 5% Claude, 12% Gemini), which the tie-extended likelihood absorbs. The experiment covers three frontier models; smaller models' self-reports may behave differently.

L Scoring-Rule Validity: Variance, Refusals, and Embeddings

This appendix collects the evidence that the scoring rule—LLM judge, multiplicative coverage×accuracy, panel mean—measures the entity rather than the panel, and that no simpler rule would do (Section 4).

L.1 The score measures the entity, not the panel

Decomposing the total sum of squares over all records (Table 18), the entity factor explains more than three times the variance the model factor does. Model identity contributes a minority share, and the per-model generosity differences that produce it (frontier reasoning models at the generous end, small RL-tuned-against-hallucination models at the strict end) are tabulated in full in the released per-model statistics.

Table 18. Variance decomposition over 211,603 records (5,719 entities × 37 models). Entity and cohort are nested, so their rows overlap rather than partitioning the total; each row is the share of total SS that the factor independently explains.

Source	SS	% of total
Entity	9,417	**35.8%**
Model	2,873	10.9%
Cohort	4,425	16.8%
Residual (interactions)	14,035	53.3%

L.2 Refusal structure validates the multiplicative rule

The refusal rate—the fraction of the panel answering "unknown"—inverts NameRank almost perfectly at cohort level (full cohort table with refusal rates in the released cohort specification): the silent cohorts draw refusal rates above 50%, the universal cohorts near zero. Models honestly refuse on low-recognition cohorts rather than hallucinating, which is the threshold-cleanliness property surfacing in raw response behavior.

The per-model refusal signature shows why the multiplicative rule is load-bearing. The high-refusal cluster is dominated by RL-tuned-against-hallucination models; the zero-refusal cluster consists of *fluent hallucinators* that always produce a confident-looking biography (released per-model statistics). The accuracy axis is what disciplines the latter: a wrong-person bio scores high coverage but near-zero accuracy, so its product contributes nothing. A coverage-only or refusal-only metric would systematically over-credit the fluent-hallucinator cluster.

L.3 Embedding similarity cannot replace the judge

Across 175,397 non-refusal records, the correlation between judge score and embedding similarity is only 0.152, and embedding similarity is essentially flat across judge-score buckets (Table 19). Embeddings measure topical similarity, which is high even for hallucinated bios: a fabricated biography of a CS researcher still embeds close to a gold answer about that researcher's actual subfield. The most extreme disagreements—judge 0.0, embedding ≈0.97—are confident, entirely fabricated bios of Chinese-name long-tail researchers, exactly the fluent-hallucination case the judge exists to catch (a worked example is in the repository documentation). An embedding-only NameRank would credit these as recognized.

Table 19. Mean embedding similarity by judge-score bucket (175,397 non-refusal records). Embedding similarity does not separate unrecognized from recognized entities.

Judge score bucket	n	Mean embedding sim.
0.0–0.1	23,621	0.819
0.1–0.3	14,358	0.798
0.3–0.5	26,291	0.801
0.5–0.7	29,616	0.818
0.7–0.9	52,534	0.843
0.9–1.0	28,977	0.829

L.4 Per-cohort within-entity disagreement

The within-entity model-disagreement (standard deviation across the 37-model panel for each entity, averaged within cohort) ranges over an order of magnitude, from 0.404 (NOI China gold) down to 0.113 (online courses); Figure 9 gives the full ladder.

Interpretation. High within-entity σ cohorts (NOI/ICPC/CMO/IMO/long-tail-OpenAlex) exhibit genuine corpus-pocket-specific recognition: different models trained on different fractions of available corpus see different fractions of the entity's coverage. Low within-entity σ cohorts (online courses, conferences, named methods, books) have universal corpus presence: all models agree because the entity name appears in nearly every training-data source.

The variance pattern is itself a probe of corpus-pocket distribution: cohorts with high σ are the ones where "which model do you ask?" changes the answer.

M Prompt-Paraphrase Sensitivity

The headline metric uses a single fixed probe template (Section 2.1). The *ordering* it produces is robust to wording: panel-mean NameRank correlates at $r = 0.93$–0.98 across template pairs, the cohort ladder is invariant under all four templates—including the working-researcher-above-IMO-gold relationship that underlies the credential treadmill—and template is by far the smallest main effect in the variance decomposition (Table 20). The *absolute level* is not: the original template draws fuller responses, and hence systematically higher coverage, than terser paraphrases, and within a single (entity, model) cell the wording sensitivity is comparable to the model-identity sensitivity. What rescues the metric is the panel mean, whose cross-template variance is more than an order of magnitude smaller than any single model's. NameRank values should therefore be read as template-conditional, and cross-study comparisons must hold the probe wording fixed.

We re-probed a cohort-stratified 200-entity subset (30 each from IMO gold, OpenAlex long-tail researchers, CS faculty, Writers, OSS projects, GPT-5 system-card authors; 10 each from Research papers and Filmmakers) on the full 37-model panel under three paraphrased probe templates (T1–T3) alongside the original (T0), for 29,600 records. The four templates:

```
T0: Tell me what you know about {name}, who is {context}. If you do
```

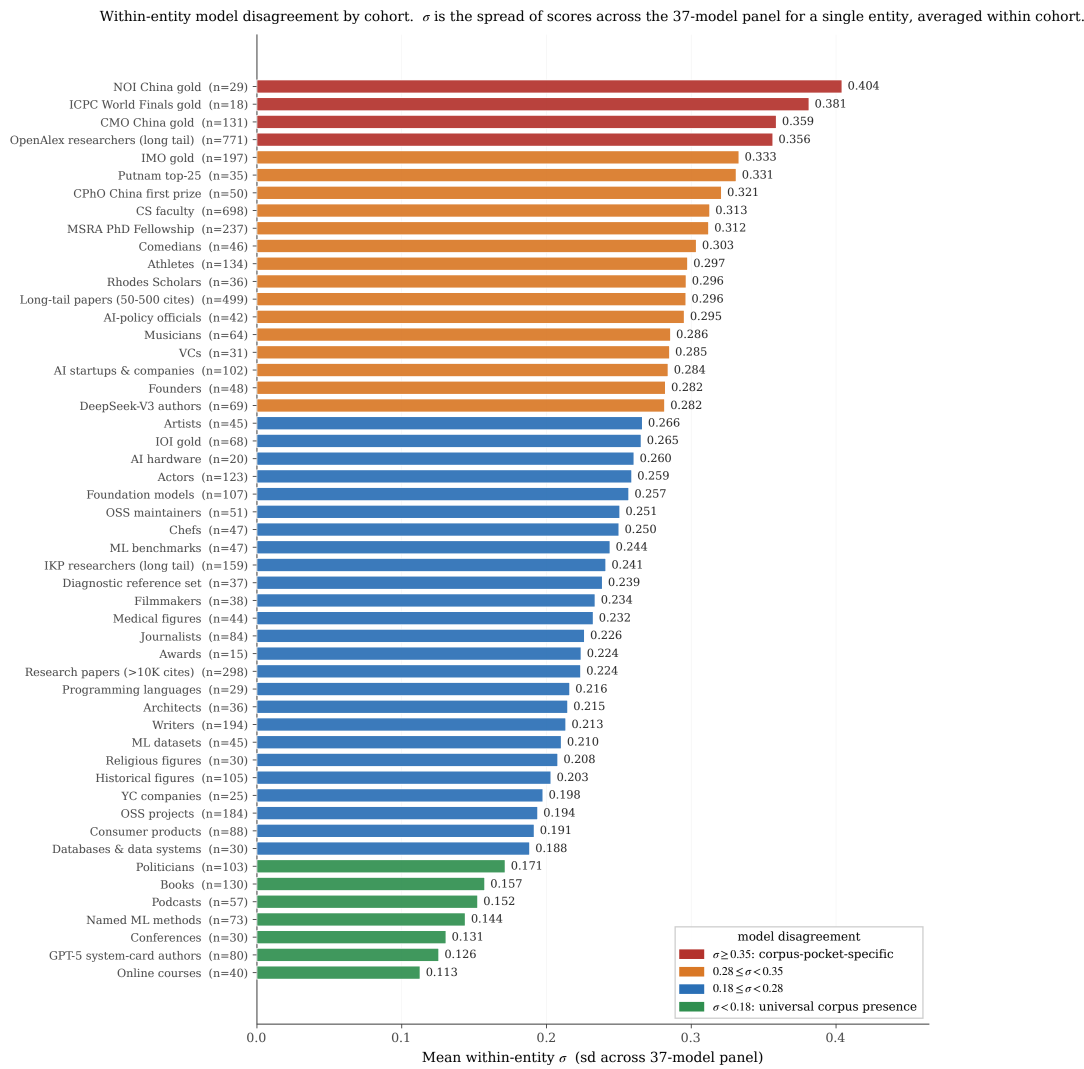

Figure 9. Full within-entity σ ladder across all cohorts with $n \geq 10$. Color bands group cohorts by the four σ regimes (universal $\to$ corpus-pocket-specific). The high-σ end is dominated by competition-credential and long-tail academic cohorts whose recognition is genuinely corpus-fragment-specific; the low-σ end is dominated by entities whose names appear in nearly every training source (online courses, conferences, named methods, books).

```
    not recognize this entity, respond with "unknown". Limit your
    response to about 150 words.
T1: What can you tell me about {name}? They are {context}. Reply with
    only "unknown" if you are not familiar with this entity. Aim for
    about 150 words.
T2: Please share what you know about {name}, {context}. If unknown to
    you, answer with just "unknown". Keep it under ~150 words.
T3: Briefly describe {name}, who is {context}. Respond "unknown" if
    you have no information. Around 150 words please.
```

Table 20. Additive variance decomposition over the 29,600 prompt-sensitivity records (entity, model, template, cohort as main effects). Template is the smallest main effect. The model share exceeds the full-panel 10.9% (Section 4) because this cohort-stratified subset over-represents silent and universal cohorts, compressing the entity-vs-model contrast; the valid within-experiment comparison is template (1.78%) vs model (26.5%).

Factor	SS	% of total
Entity	985.3	24.19%
Model	1079.9	26.51%
Cohort (entity-implied)	551.9	13.55%
Template	**72.4**	**1.78%**
Residual (after entity+model+template)	1936.3	47.53%

Table 21. Cohort-mean NameRank under each template (sorted by T0). The ordering is preserved; T0 is systematically higher. Pairwise panel-mean Pearson across all six template pairs is 0.927–0.977 (lowest: T0–T3); within-(entity, model) cross-template σ has median 0.109; the panel-mean variance across templates is 0.042$\times$ the single-template cross-model variance.

Cohort	T0	T1	T2	T3
Filmmakers	0.761	0.506	0.513	0.532
Writers	0.597	0.388	0.403	0.405
Research papers	0.581	0.391	0.394	0.401
OSS projects	0.504	0.334	0.335	0.331
OpenAlex long-tail researchers	0.465	0.362	0.341	0.455
CS faculty	0.384	0.300	0.295	0.319
IMO gold	0.356	0.290	0.260	0.302
GPT-5 system-card authors	0.047	0.063	0.027	0.070
Panel template mean	**0.420**	**0.305**	**0.294**	**0.329**

N Training-Cutoff Gradient: The Silent Zone Is Intrinsic

A natural worry about the silent zone (Section 2.9) is that it reflects *corpus timing* rather than *intrinsic obscurity*: perhaps low-NameRank entities are merely too recent for current models to have ingested, and a future fleet would lift them. Two contemporaneously-published cohorts test this directly. The DeepSeek-V3 author cohort became nameable in any indexable corpus only when the technical report was posted (December 2024); the GPT-5 system-card author cohort only at the system card's publication (August 2025). The panel's training cutoffs straddle both dates (released with the data; cutoffs span 2023-10 to 2026-03), so a model whose cutoff predates an emergence date *cannot* have ingested the document.

The result (Figure 10) separates two effects. First, a model-vintage confound dominates the cutoff axis: *every* cohort's recognition rises with training cutoff at roughly 0.10–0.20 per year, including cohorts (containing, e.g., Geoffrey Hinton) nameable long before any cutoff—capability and willingness-to-commit drift, not corpus growth. Second, once that drift is netted out by matched difference-in-differences against stable control cohorts, no corpus-timing signal remains: the DeepSeek-V3 cohort's pre/post-emergence jump matches the controls', and the GPT-5 system-card cohort goes *more* silent in post-cutoff models, whose hallucination-averse tuning correctly refuses on a name appearing only in a 486-person contributor list.

Two implications follow. *Recognition is not ingestion*: presence in the training corpus is necessary but far from sufficient, so crossing a cutoff does not manufacture recognition for an entity below the corpus-density threshold. And *NameRank is not comparable across model vintages*: the ∼0.13/yr drift means longitudinal claims—including the predictions in Section 5.9—must be made on within-run *relative* gaps, never absolute levels. The thresholded behavior has a mechanistic basis in how transformers store and retrieve facts: feed-forward layers act as key–value memories for factual associations [Geva et al., 2021], with identifiable knowledge neurons per fact [Dai et al., 2022, Meng et al., 2022]; pretrained models function as implicit knowledge bases [Petroni et al., 2019] whose factual capacity scales with model size [Roberts et al., 2020]; accuracy on rare facts scales log-linearly with fact popularity [Kandpal et al., 2023, Mallen et al., 2023]; and popular knowledge actively suppresses less popular knowledge [Zhang et al., 2025]. Below the corpus-density threshold, silence rather than noise is what this retrieval stack predicts—which is what we observe.

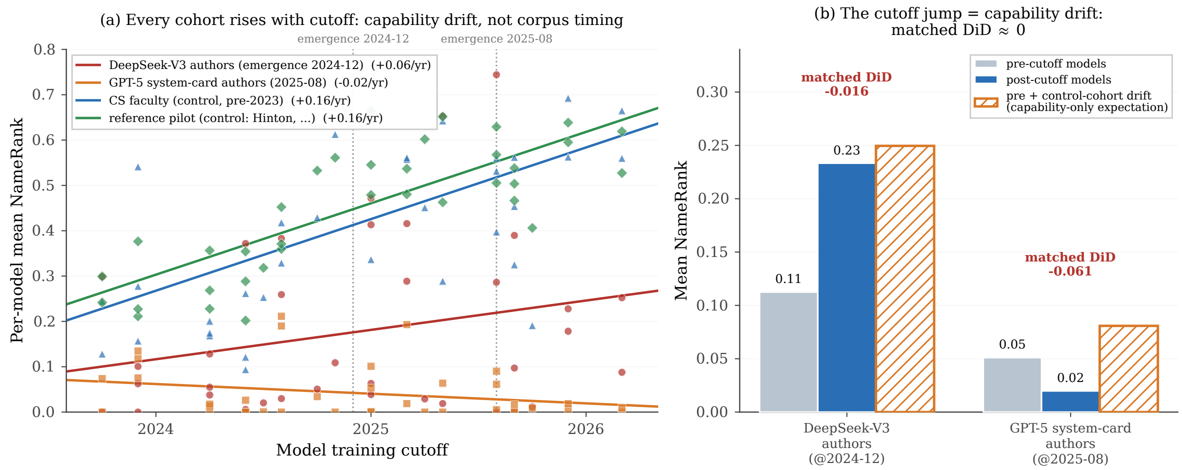

Figure 10. Training-cutoff gradient. **(a)** Per-model mean NameRank vs. training cutoff for two first-appearance "recency" cohorts (DeepSeek-V3 authors, emergence 2024-12; GPT-5 system-card authors, 2025-08) and two stable controls nameable since well before any cutoff. *Every* cohort rises with cutoff at ∼0.10–0.20/yr—a model-capability confound, not corpus timing (one control cohort contains Geoffrey Hinton). **(b)** Natural experiment: the recency cohorts' pre/post-emergence jump matches the same-split jump on the stable controls (hatched "capability-only expectation" bar), so the matched difference-in-differences is ≈0. Crossing a model's training cutoff is necessary but not sufficient for recognition.

Table 22. Natural experiment: per-model mean recognition split by whether the model's training cutoff predates (pre) or postdates (post) each cohort's emergence date. The raw jump is decomposed by a matched difference-in-differences (DiD) that subtracts the same-split jump averaged over three stable controls (CS faculty, OpenAlex long-tail researchers, IMO gold), which nets out the cross-vintage capability drift. Both matched DiDs are ≤ 0.

Cohort	pre (refusal)	post (refusal)	raw jump	control	DiD
DeepSeek-V3 authors	0.113 (0.38)	0.233 (0.45)	+0.121	+0.137	−**0.016**
GPT-5 system-card authors	0.051 (0.49)	0.020 (0.79)	−0.031	+0.030	−**0.061**

Table 23. Selected per-cohort slopes of per-model mean NameRank on training cutoff (NameRank per year of cutoff), with Pearson r over the 37 models. *Every* established cohort rises with cutoff at $\sim$0.10–0.20/yr—a model-capability/generosity confound, since these entities were nameable long before any cutoff. The two recency cohorts are *not* steeper; once the drift is removed (Table 22) no corpus-timing signal remains.

Cohort	slope (/yr)	Pearson r
Comedians	+0.196	+0.66
Foundation models	+0.171	+0.76
CS faculty (control)	+0.158	+0.57
Reference set (control; incl. Hinton)	+0.157	+0.77
Research papers	+0.125	+0.74
OpenAlex long-tail researchers (control)	+0.094	+0.32
DeepSeek-V3 authors (recency)	+0.065	+0.23
IMO gold	+0.043	+0.15
GPT-5 system-card authors (recency)	−0.021	−0.25

O Cross-Judge Robustness

NameRank uses a single LLM judge (Section 2.4). To test whether the findings depend on that choice, we re-judged a 615-record stratified sample—spanning the diagnostic reference set, CS faculty, OpenAlex researchers, IMO gold medalists, GPT-5 system-card authors, writers, and OSS projects—with two additional judges from different families (Claude Opus 4.6 and GPT-5.5), holding gold answers and judge prompt fixed.

The three judges agree closely: pairwise Pearson 0.87–0.93 across the full sample (Table 24), 0.95 on the reference-set anchors, and the cohort ladder—including the credential-treadmill ordering—is preserved under all three. A self-preference exists but is small once response capability is controlled (Table 25): Gemini awards its own family a residual +0.062 and GPT-5.5 +0.041, while Claude shows none. This offset shifts per-vendor score tables (released per-model statistics) but not the entity- and cohort-level comparisons that constitute the findings, which average over the full panel and are invariant to the judge. For future runs, Claude is the least self-favoring single judge, and a three-judge mean removes the offset entirely.

Table 24. Pairwise Pearson correlation between judges, by cohort. Agreement is high everywhere except the $n = 15$ OSS projects cell (generic-description disagreement). The Reference set anchors—including its four Chinese-name entities—show the tightest agreement, refuting a "Chinese name $\rightarrow$ judge disagreement" hypothesis.

Subset	n	Gem–Claude	Gem–GPT-5	Claude–GPT-5
ALL	615	0.868	0.893	0.934
Reference set	185	0.952	0.941	0.972
(Chinese-name subset)	74	0.957	0.945	0.970
OpenAlex long-tail researchers	100	0.855	0.920	0.935
CS faculty	100	0.712	0.894	0.781
IMO gold	100	0.785	0.808	0.924
GPT-5 system-card authors	100	0.781	0.917	0.830
Writers	15	0.872	0.853	0.878
OSS projects	15	0.103	0.455	0.486

Table 25. Capability-controlled in-family lift. For each response, a judge's score is compared to the mean of the *other two* judges on that same response; the lift is the own-family residual minus the other-family residual. Gemini and GPT-5 modestly favor their own family; Claude does not.

Judge	In-family residual	Other-family residual	Lift
Gemini	+0.014	−0.048	+**0.062**
GPT-5	+0.060	+0.019	+**0.041**
Claude	+0.012	+0.022	−**0.010**

Cohort-mean NameRank is stable across judges, and the silent→discriminative→universal ladder is preserved under all three: on this stratified 615-record subsample (whose per-cohort means differ from the full-run means by construction), the GPT-5 system-card cohort scores 0.087/0.165/0.123 under Gemini/Claude/GPT-5, and the OpenAlex long-tail cohort 0.669/0.630/0.631. The headline findings are not a single-judge artifact.

P Synthetic-Null Floor

To establish what NameRank assigns to a name absent from every pretraining corpus, we ran 30 entirely fictional entities—spanning the archetypes the paper measures, with verified-ungoogleable names, cohort-matched contexts, and matched gold answers—through the full pipeline. Table 26 gives the per-archetype floor: near 0.04 for archetypes with dense, specific golds (essentially the main-run silent-zone level), but near 0.20 for the synthetic IMO-gold archetype, whose gold is mostly cohort boilerplate (year, country, medal) that a model can partly satisfy from the disambiguating context alone without recognizing the nonexistent person.

This calibration *sharpens* the credential treadmill rather than threatening it. The olympiad cohorts carry the ∼0.20 boilerplate-gold floor, so IMO gold clears its floor by far less than the OpenAlex baseline clears its ∼0.04 dense-gold floor—the floor-adjusted gap is wider than the raw gap. The corollary caveat: the lowest credential cohorts sit at or below their floor and should be read as silent rather than weakly recognized; differences smaller than the relevant floor are not evidence of recognition.

The floor is concentrated in small open-weights fluent hallucinators: gemma-3-4b (0.293), gemma-3-12b (0.287), and llama-4-maverick (0.259) lead, while 15 of the 37 models score the synthetics at exactly 0 (including grok-4, kimi-k2, qwen3-235b, mistral-large, gemini-3.1-pro, and gpt-5.3, the last two by refusing on 100% of synthetics). Recommended use: treat ∼ 0.04 as the noise floor for dense-gold cohorts and ∼ 0.20 for short-boilerplate-gold cohorts; differences below the relevant floor are not evidence of recognition.

Q Confound Checks: Gold Length, Context, Wikipedia, and Attention

Four alternative explanations for the headline findings were tested and rejected: that the credential gap is a gold-answer-length artifact, that the disambiguating context leaks the answer, that NameRank reduces to Wikipedia presence, and that NameRank reduces to a web-attention ranking.

Gold-answer length. Within-cohort regressions of NameRank on gold-answer length give $R^2 < 0.07$ for IMO (0.067), IOI (0.018), and MSRA (0.006); the high-R^2 cohorts are fame-graded

Table 26. Synthetic-null mean NameRank by archetype. Dense-gold archetypes (founder, CS faculty) floor near the main-run silent level; the short-boilerplate-gold IMO archetype floors much higher, because its generic year+country+medal gold is partly satisfiable from the disambiguating context without recognizing the (nonexistent) person.

Synthetic archetype	n	Mean NameRank	Refusal
IMO gold (boilerplate gold)	6	**0.199**	21%
Chef	1	0.123	27%
Journalist	1	0.092	27%
Podcast	1	0.075	32%
OSS project	4	0.065	39%
Musician	1	0.057	41%
CS faculty	10	**0.027**	45%
Founders	6	**0.012**	47%
All synthetics	30	0.072	—
All except IMO archetype	24	**0.040**	—

mid-tier categories (actor 0.95, writer 0.88) where longer golds are an outcome of fame, not a measurement artifact. The credential-vs-OpenAlex ordering survives a pooled fixed-effects length adjustment (which deepens the gap: pooled slope $b_{\text{chars}} = -1.1\times10^{-3}$) and a within-OpenAlex-slope adjustment. A strict $|\Delta\text{chars}| \leq 50$ matched-pairs test yields $n = 0$ pairs: IMO golds span 359–392 characters, OpenAlex golds 662–789, with no overlap. The closest feasible match (IMO vs. CMO, which do overlap in length) preserves the within-credential ordering.

Disambiguating context. A 288-entity subset re-probed with the specific context replaced by a minimal generic descriptor:

Table 27. Minimal-context ablation. Stripping the specific disambiguating context to a generic descriptor drops absolute NameRank substantially but preserves the cross-entity gradient.

Comparison	Specific context	Minimal context
OpenAlex long-tail researchers (mean)	0.446	0.077
IMO gold (mean)	0.345	0.039
Stanford mean	0.565	0.355
Tsinghua mean	0.258	0.062
Stanford−Tsinghua gap	**0.308**	**0.293** (−4.8%)

Absolute levels fall sharply (the model cannot identify which researcher without the context), but the Stanford–Tsinghua institutional gap shrinks by only $\sim 5\%$: context anchors disambiguation, it does not leak the gold answer.

Wikipedia presence. On the $n = 771$ long-tail cohort, a binary has-Wikipedia flag explains $R^2 = 0.022$ of NameRank (log-sitelinks and log-pageviews add +0.007 and 0.000 on top), versus $R^2 = 0.398$ for log(h-index) alone. Adding log(h-index) to the full Wikipedia block lifts R^2 from 0.029 to 0.405—a marginal R^2 of 0.376. Across the full 5,719-entity cohort, Wikipedia presence explains $\sim 8\%$ of variance and cohort identity $\sim 47\%$. Wikipedia presence accounts for only $\sim 1/4$ of the Stanford–Tsinghua gap. NameRank tracks bibliometric productivity, not a Wikipedia-yes/no flag.

Web attention. Attention metrics place entities on a single cross-domain axis, so we test the strongest freely available one directly: 24-month en-Wikipedia article pageviews (2023-07 through 2025-06; Wikimedia REST, user traffic) for every entity whose article the strict-disambiguation lookup above resolves—1,617 entities after excluding 48 whose articles were created after the window and 16 disambiguation false positives. Figure 11 summarizes three results. First, the metric is undefined for the 71% of entities with no article, whose NameRank spans 0.16–0.68 (p10–p90)—the entire discriminative zone, where the credential, faculty, and long-tail findings live. Second, where defined, log-pageviews explain $R^2 = 0.06$ of NameRank overall and 0.10 within cohort; on the 58 long-tail researchers where both predictors exist, the pageview correlation is *negative* ($r = -0.33$) while $\log(h\text{-index})$ alone reaches $R^2 = 0.39$. Third, the within-cohort correlation carries a sign structure: positive for celebrity cohorts (actors +0.81, athletes +0.77, writers +0.62), negative for technical ones (OSS projects −0.54, databases −0.44, programming languages −0.35). The divergence is informative rather than noise: Chainer (6.4K views over the window; NameRank 0.94) and Theano (26K; 0.89) are corpus-saturated tools nobody looks up anymore, while Stable Diffusion (1.3M views; 0.38) draws enormous current traffic against a shallower accumulated corpus. Pageviews measure the *flow* of current curiosity; NameRank measures the accumulated *stock* of name-bearing text as absorbed at training time (Section 5.3)—related axes, not the same axis. The excluded post-window articles make the point from the other side: the vLLM and Ollama articles were created only in 2026, yet both entities already score in the discriminative band—recognition no pageview record could have predicted.

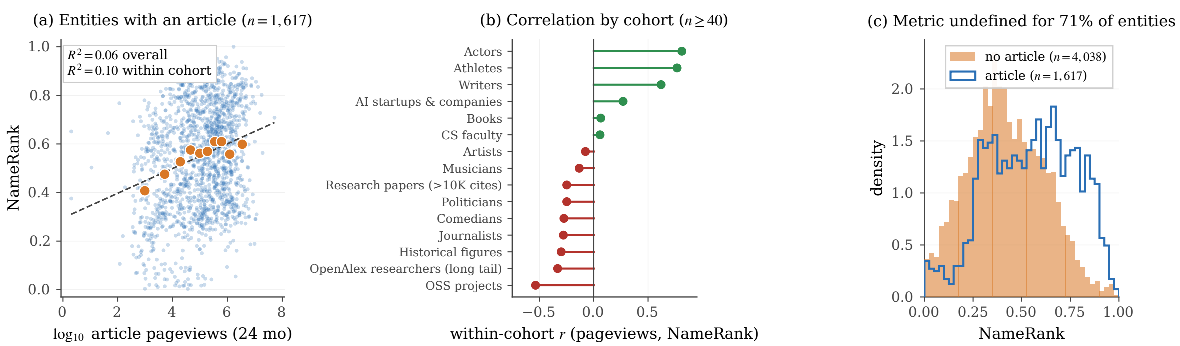

Figure 11. NameRank is not Wikipedia-pageview rank. **(a)** NameRank against $\log_{10}$ 24-month article pageviews for the 1,617 entities with a resolvable article (small dots: entities; large markers: decile means): $R^2 = 0.06$ overall, 0.10 within cohort. **(b)** Within-cohort correlation for cohorts with $n \geq 40$ matched entities: positive only for celebrity cohorts, negative for technical and production cohorts—current attention flow vs. accumulated corpus stock. **(c)** NameRank distribution for entities with and without an article: the attention metric is undefined for the 71% without one, whose scores span the discriminative zone.

R Gender Gap by Cohort

First-name-inferred gender (via `gender-guesser`; ambiguous and unisex names dropped) for the person cohorts, restricted to cohorts with ≥ 5 man- and woman-coded names. Table 28 reports the man-minus-woman NameRank gap.

The within-researcher-cohort gap (+0.027 to +0.054, ∼6–8% relative attenuation) is roughly 4× smaller than the ∼2× East-Asian-surname attenuation reported by Li [2026], but repro-

Table 28. Man-minus-woman NameRank gap by cohort. The controlled researcher cohorts (top) show a small reproducible man-coded advantage that survives institution- and h-index-matching. The cross-profession cohorts (bottom) show much larger raw gaps that reflect real-world fame differences in the sampled individuals, not a NameRank artifact—hence the bias claim is restricted to the controlled researcher comparison.

Cohort	n_{man}	n_{woman}	Δ (man−woman)	p
CS faculty	354	97	+0.038	0.014
institution-matched (60 pairs)	—	—	+0.054	0.042
OpenAlex long-tail researchers	388	76	+0.031	0.182
h-index-decile-adjusted	—	—	+0.032	—
IKP long-tail researchers	56	22	+0.044	0.015
Actors	38	63	+0.341	$< 10^{-3}$
Athletes	64	32	+0.254	$< 10^{-3}$
Writers	62	77	+0.041	0.41
Comedians	18	17	−0.045	0.23

ducible across three researcher cohorts and robust to dropping medium-confidence gender labels. The large cross-profession gaps (actor, athlete) are excluded from the bias claim because the sampled men and women in those cohorts differ in actual fame.

Name-coding caveat. `gender-guesser` infers gender from first names against a predominantly Western-name dictionary; it returns "unknown" or "andy" (androgynous) for most Chinese, Korean, and other romanized East-Asian given names, which are then dropped. The gendered sub-sample is therefore enriched for Western names, and the residual gender gap is partly entangled with the East-Asian-surname attenuation it sits beside—a name that codes as gender-unknown also tends to be one of the low-recognition non-Anglophone names. The n_{woman} counts (e.g. 76–97 across the researcher cohorts) reflect this attrition. We report the gap as an inherited corpus-and-judge property restricted to the controlled within-cohort comparison, not as a clean estimate of a pure gender effect net of name origin; disentangling the two would require a gender labeling that is accurate on CJK names (e.g. manual coding or a name-origin-aware classifier).

S Limitations Summary Table

Table 29. Compact summary of methodological caveats, their direction (could overstate or understate findings), and mitigations available for future runs.

Caveat	**Direction**	**Mitigation**
Single-probe-per-entity; absolute levels are wording-conditional (Appendix M)	Shifts cardinal levels ∼0.10–0.20; rankings unaffected	*Tested:* four-template re-probe shows ordering robust ($r = 0.93$–0.98); report values as T0-conditional and hold wording fixed across studies
Judge family-bias (Appendix O)	Gemini-judging-Gemini lift +0.062 inflates per-vendor score tables; cross-entity findings unaffected	*Tested:* 3-judge re-grade ($n = 615$); report per-vendor tables under Claude or as a 3-judge mean
Cohort labels not fully orthogonal	Overcounts entities in cross-cohort comparisons	De-duplicate by name within long-tail academic categories
Chinese-prompt context fields remained English	Understates cross-language effect	Pure-Chinese prompts for the full 240-entity sub-run
Single judge on the full run (Gemini 3 Flash Preview)	Family-specific scoring style on per-vendor tables	*Tested:* non-Gemini replication (Claude, GPT-5) on $n = 615$; a full multi-judge consensus run would remove the offset
Cross-section, not longitudinal	Cannot infer name-propagation rates	Re-run at each major model-release cycle; release each snapshot dated
Scores not comparable across model vintages (Appendix N)	∼0.13/yr capability drift inflates any cross-fleet level comparison	*Tested:* silent zone is intrinsic, not timing (matched DiD ≈0); make longitudinal claims on within-run relative gaps only
Researcher mobility confound in country gradient	Inflates US averages, deflates China averages	Re-classify by country-of-PhD or country-of-doctorate-training as alternative slicing
Tsinghua/Peking small samples ($n = 4$)	Institution-level reads noisy at the bottom	Expand sample by drawing from CCF rank-A faculty rosters
Tianshou case study sample of 1 for per-response mediation analysis	Cannot generalize the +0.36 score lift cleanly	Extend per-response mediation analysis to all 11 verified (creator, artifact) pairs
Inherited gender bias (Appendix R)	Man-coded researcher names score +0.03–0.05 over woman-coded	*Measured, not corrected:* report alongside the East-Asian-surname effect; restrict to controlled within-cohort comparisons
Short-boilerplate-gold cohorts have an elevated noise floor (Appendix P)	Olympiad-credential scores up to ∼ 0.20 are partly context-prior leakage	Read scores against the synthetic floor (∼ 0.04 dense-gold, ∼ 0.20 short-gold); the floor widens, not narrows, the credential gap
Cohort-base-rate leakage in the echo-discounting judge (Appendix D)	The judge discounts facts restated from the probe context but not facts predictable from the cohort trajectory; where a gold's distinguishing facts *are* the typical path (e.g. most NOI golds attend Tsinghua/the Yao Class), a model—even a fluent hallucinator—earns coverage credit by producing the base-rate story without recognizing the individual, spuriously lifting generic-trajectory entities and leaving idiosyncratic ones (named startups, tools, viral events) conservative	*Not corrected:* extend the judge to also discount facts inferable from the disambiguating cohort, crediting only individual-identifying content; report the residual tail ranking
Institution variance share inflated by category count (Section 3.5.2)	Naive $R^2 = 0.54$ counts 319 dummies on 698 points; true share ∼16%	*Reported:* report adjusted R^2/ω^2/ICC (∼0.16) and the large-cell Stanford–Tsinghua contrast; country-level aggregation is not inflated
Context-injection lift includes a coverage-echo component	Overstates the named-artifact-amplification	Re-judge configuration B against an artifact-redacted gold crediting only